```
import pandas as pd

#load the data
url = "https://raw.githubusercontent.com/saugat977/DAV-6150/refs/heads/main/M6/Project1_Data.csv"
df = pd.read_csv(url)
print(df.shape)
df.head(20)
```

```
# Columns to convert to numeric (all _cnt and _pct columns)
numeric_cols = [
    'enroll_cnt', 'grad_cnt', 'local_cnt', 'reg_cnt', 'reg_adv_cnt',
    'non_diploma_credential_cnt', 'still_enr_cnt', 'ged_cnt', 'dropout_cnt',
    'grad_pct', 'local_pct', 'reg_pct', 'reg_adv_pct',
    'non_diploma_credential_pct', 'still_enr_pct', 'ged_pct', 'dropout_pct'
]
# Convert count columns (replace '-' with None, convert to float)
count_cols = [col for col in numeric_cols if col.endswith('_cnt')]
for col in count_cols:
    df[col] = df[col].replace('-', None).astype('float')

# Convert percentage columns (replace '-' with None, remove '%', convert to float)
pct_cols = [col for col in numeric_cols if col.endswith('_pct')]
for col in pct_cols:
    df[col] = df[col].replace('-', None).str.rstrip('%').astype('float')

# Verify
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 73152 entries, 0 to 73151
Data columns (total 29 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   report_school_year           73152 non-null  object
 1   aggregation_index            73152 non-null  int64
 2   aggregation_type             73152 non-null  object
 3   aggregation_name             73152 non-null  object
 4   nrc_code                     73152 non-null  int64
 5   nrc_desc                     73152 non-null  object
 6   county_code                  73152 non-null  int64
 7   county_name                  73152 non-null  object
 8   nyc_ind                      73152 non-null  int64
 9   membership_desc              73152 non-null  object
 10  subgroup_code                73152 non-null  int64
 11  subgroup_name                73152 non-null  object
 12  enroll_cnt                   39674 non-null  float64
 13  grad_cnt                     39674 non-null  float64
 14  grad_pct                     39674 non-null  float64
 15  local_cnt                    39674 non-null  float64
 16  local_pct                    39674 non-null  float64
 17  reg_cnt                      39674 non-null  float64
 18  reg_pct                      39674 non-null  float64
 19  reg_adv_cnt                  39674 non-null  float64
 20  reg_adv_pct                  39674 non-null  float64
 21  non_diploma_credential_cnt   39674 non-null  float64
 22  non_diploma_credential_pct   39674 non-null  float64
 23  still_enr_cnt                39674 non-null  float64
 24  still_enr_pct                39674 non-null  float64
 25  ged_cnt                      39674 non-null  float64
 26  ged_pct                      39674 non-null  float64
 27  dropout_cnt                  39674 non-null  float64
```

```
 28  dropout_pct                39674 non-null  float64
(73152, 29)
dtypes: float64(17), int64(5), object(7)
memory usage: 16.2+ MB
```

	report_school_year	aggregation_index	aggregation_type	aggregation_name	nrc_code	nrc_desc	county_code	county_nam
0	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
1	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
2	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
3	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
4	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
5	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
6	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
7	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
8	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
11	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
12	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
15	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN
16	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN

EDA

report_school_year

```
# report_school_year
df['report_school_year'].describe()
```

	report_school_year
count	73152
unique	1
top	2018-19
freq	73152

dtype: object

Here, all the values are 2018-19 so we can exclude this column as it will not impact our model.

aggregation_index

```
#aggregation_index
df['aggregation_index'].describe()
```

	aggregation_index
count	73152.0
mean	3.0
std	0.0
min	3.0
25%	3.0
50%	3.0
75%	3.0
max	3.0

dtype: float64

Here, all the values are 3.0 and we can exclude this column too like for report_school_year

aggregation_type

```
#aggregation_type
df['aggregation_type'].describe()
```

	aggregation_type
count	73152
unique	1
top	District
freq	73152

dtype: object

Same as report_school_year, we will be excluding this column

Aggregation_name

12	2018-19	3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN

```
#aggregation_name
school_districts = df['aggregation_name']
school_districts.describe()
```

	aggregation_name
count	73152
unique	680
top	KINGSTON CITY SCHOOL DISTRICT
freq	138

20 rows × 29 columns

		3	District	ALBANY CITY SCHOOL DISTRICT	3	Urban-Suburban High Needs	1	ALBAN

dtype: object

There are 680 unique nominal categorical values in this column. This column is not present on the data dictionary but can be relevant for analysis.

```
district_counts = school_districts.value_counts()
district_counts
```

aggregation_name	count
KINGSTON CITY SCHOOL DISTRICT	138
WILLIAMSVILLE CENTRAL SCHOOL DISTRICT	136
MIDDLETOWN CITY SCHOOL DISTRICT	134
ROCHESTER CITY SCHOOL DISTRICT	134
GREECE CENTRAL SCHOOL DISTRICT	134
...	...
LONG LAKE CENTRAL SCHOOL DISTRICT	70
KIRYAS JOEL VILLAGE UNION FREE SCHOOL DISTRICT	66
GREENBURGH-NORTH CASTLE UNION FREE SCHOOL DISTRICT	56
MOUNT PLEASANT-BLYTHEDALE UNION FREE SCHOOL DISTRICT	48
RANDOLPH ACADEMY UNION FREE SCHOOL DISTRICT	40

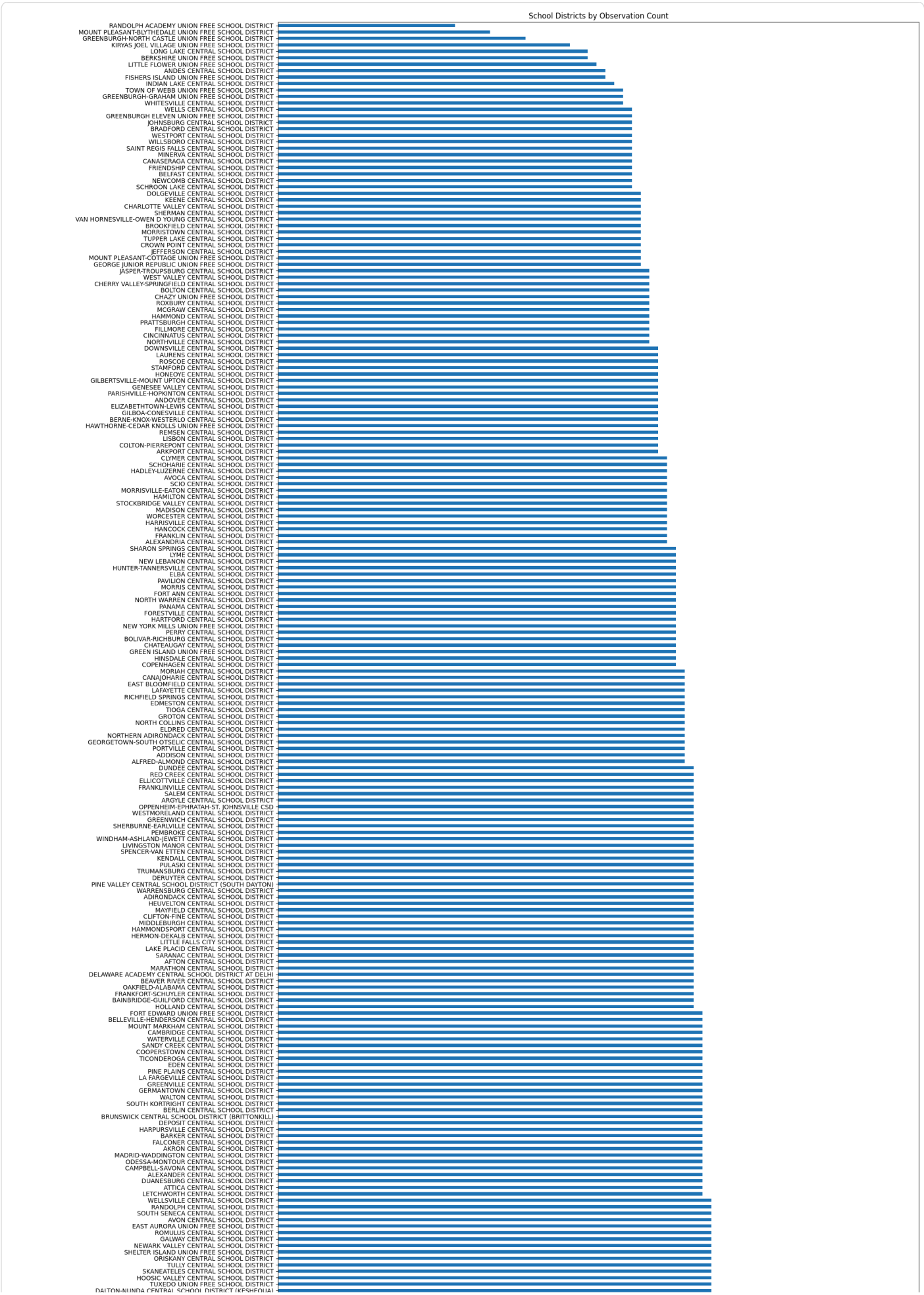
680 rows × 1 columns

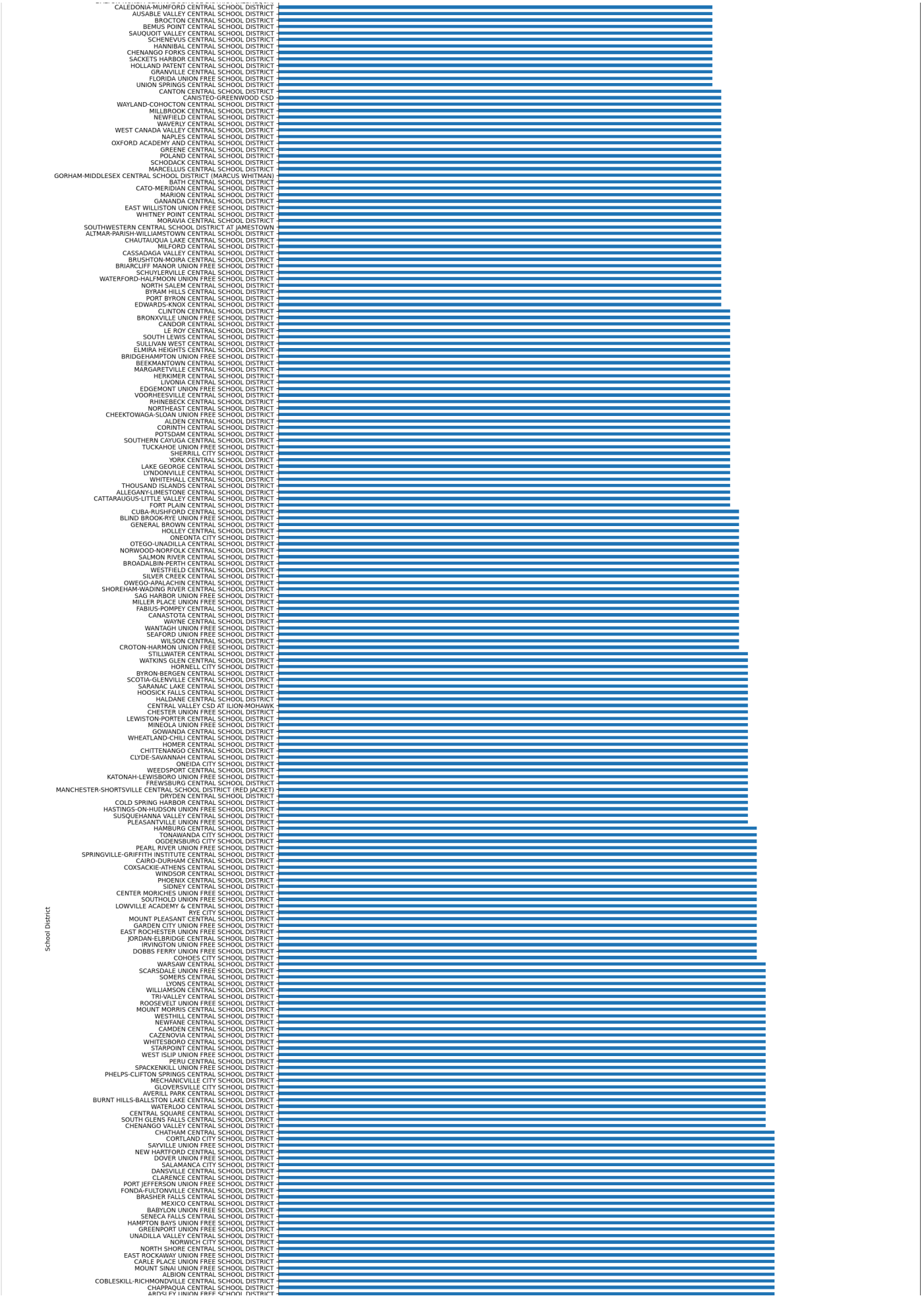
dtype: int64

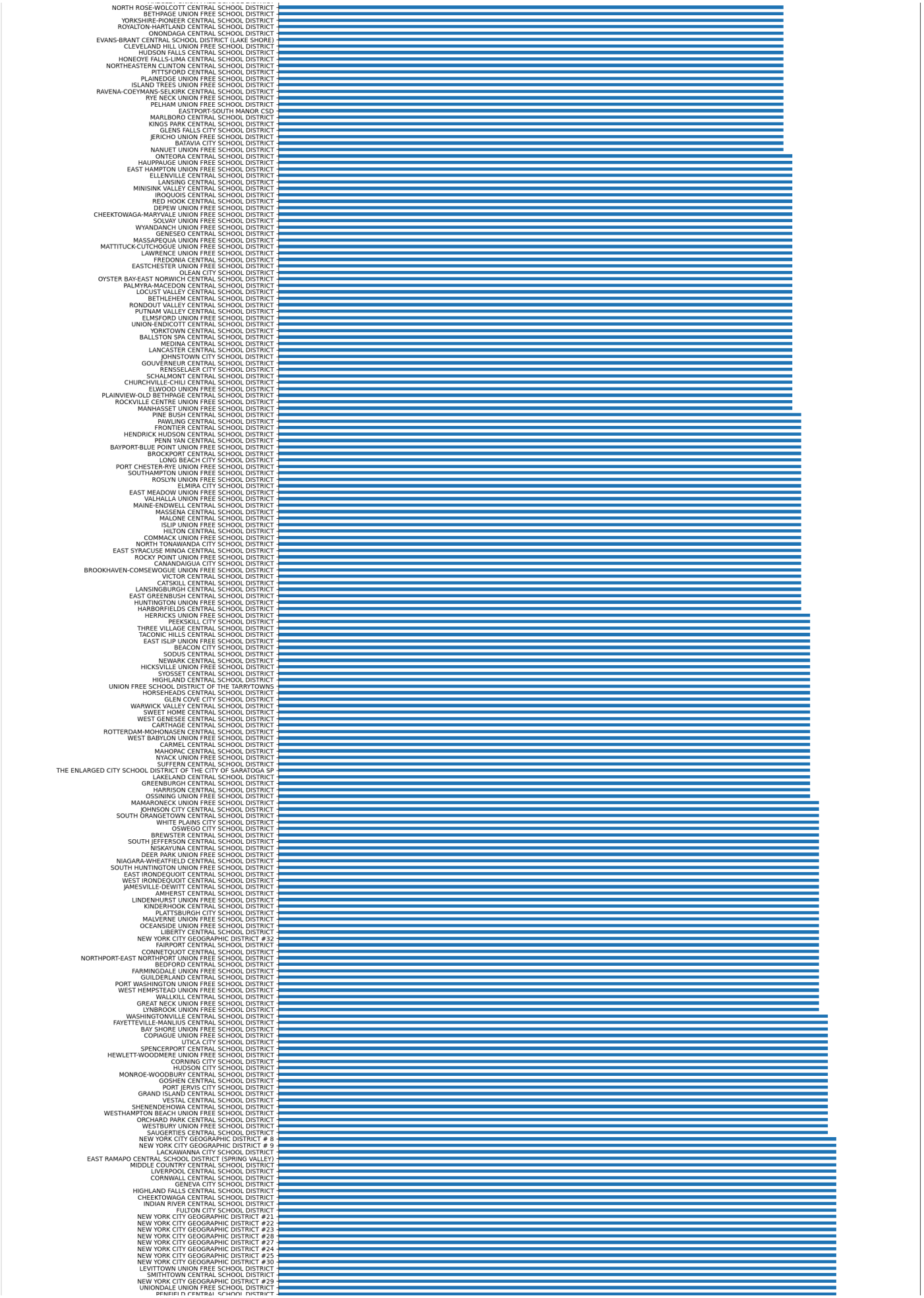
Here, the counts of districts range from 40 to 138 which is good spread for performing analysis.

```
# creating barchart to further analyse
import matplotlib.pyplot as plt

# Create bar chart
plt.figure(figsize=(20, 100))
district_counts.plot(kind='barh')
plt.xlabel('Count')
plt.ylabel('School District')
plt.title('School Districts by Observation Count')
plt.tight_layout()
plt.show()
```


School Districts by Observation Count
RANDOLPH ACADEMY UNION FREE SCHOOL DISTRICT
MOUNT PLEASANT-BLYTHEDALE UNION FREE SCHOOL DISTRICT
GREENBURGH-NORTH CASTLE UNION FREE SCHOOL DISTRICT
KIRYAS JOEL VILLAGE UNION FREE SCHOOL DISTRICT
LONG LAKE CENTRAL SCHOOL DISTRICT
BERKSHIRE UNION FREE SCHOOL DISTRICT
LITTLE FLOWER UNION FREE SCHOOL DISTRICT
ANDES CENTRAL SCHOOL DISTRICT
FISHERS ISLAND UNION FREE SCHOOL DISTRICT
INDIAN LAKE CENTRAL SCHOOL DISTRICT
TOWN OF WEBB UNION FREE SCHOOL DISTRICT
GREENBURGH-GRAHAM UNION FREE SCHOOL DISTRICT
WHITESVILLE CENTRAL SCHOOL DISTRICT
WELLS CENTRAL SCHOOL DISTRICT
GREENBURGH ELEVEN UNION FREE SCHOOL DISTRICT
JOHNSBURG CENTRAL SCHOOL DISTRICT
BRADFORD CENTRAL SCHOOL DISTRICT
WESTPORT CENTRAL SCHOOL DISTRICT
WILLSBORO CENTRAL SCHOOL DISTRICT
SAINT REGIS FALLS CENTRAL SCHOOL DISTRICT
MINERVA CENTRAL SCHOOL DISTRICT
CANASERAGA CENTRAL SCHOOL DISTRICT
FRIENDSHIP CENTRAL SCHOOL DISTRICT
BELFAST CENTRAL SCHOOL DISTRICT
NEWCOMB CENTRAL SCHOOL DISTRICT
SCHROON LAKE CENTRAL SCHOOL DISTRICT
DOLGEVILLE CENTRAL SCHOOL DISTRICT
KEENE CENTRAL SCHOOL DISTRICT
CHARLOTTE VALLEY CENTRAL SCHOOL DISTRICT
SHERMAN CENTRAL SCHOOL DISTRICT
VAN HORNESVILLE-OWEN D YOUNG CENTRAL SCHOOL DISTRICT
BROOKFIELD CENTRAL SCHOOL DISTRICT
MORRISTOWN CENTRAL SCHOOL DISTRICT
TUPPER LAKE CENTRAL SCHOOL DISTRICT
CROWN POINT CENTRAL SCHOOL DISTRICT
JEFFERSON CENTRAL SCHOOL DISTRICT
MOUNT PLEASANT-COTTAGE UNION FREE SCHOOL DISTRICT
GEORGE JUNIOR REPUBLIC UNION FREE SCHOOL DISTRICT
JASPER-TROUPSBURG CENTRAL SCHOOL DISTRICT
WEST VALLEY CENTRAL SCHOOL DISTRICT
CHERRY VALLEY-SPRINGFIELD CENTRAL SCHOOL DISTRICT
BOLTON CENTRAL SCHOOL DISTRICT
CHAZY UNION FREE SCHOOL DISTRICT
ROXBURY CENTRAL SCHOOL DISTRICT
MCGRAW CENTRAL SCHOOL DISTRICT
HAMMOND CENTRAL SCHOOL DISTRICT
PRATTSBURGH CENTRAL SCHOOL DISTRICT
FILLMORE CENTRAL SCHOOL DISTRICT
CINCINNATUS CENTRAL SCHOOL DISTRICT
NORTHVILLE CENTRAL SCHOOL DISTRICT
DOWNSVILLE CENTRAL SCHOOL DISTRICT
LAURENS CENTRAL SCHOOL DISTRICT
ROSCOE CENTRAL SCHOOL DISTRICT
STAMFORD CENTRAL SCHOOL DISTRICT
HONEOYE CENTRAL SCHOOL DISTRICT
GILBERTSVILLE-MOUNT UPTON CENTRAL SCHOOL DISTRICT
GENESEE VALLEY CENTRAL SCHOOL DISTRICT
PARISHVILLE-HOPKINTON CENTRAL SCHOOL DISTRICT
ANDOVER CENTRAL SCHOOL DISTRICT
ELIZABETHTOWN-LEWIS CENTRAL SCHOOL DISTRICT
GILBOA-CONESVILLE CENTRAL SCHOOL DISTRICT
BERNE-KNOX-WESTERLO CENTRAL SCHOOL DISTRICT
HAWTHORNE-CEDAR KNOLLS UNION FREE SCHOOL DISTRICT
REMSEN CENTRAL SCHOOL DISTRICT
LISBON CENTRAL SCHOOL DISTRICT
COLTON-PIERREPONT CENTRAL SCHOOL DISTRICT
ARKPORT CENTRAL SCHOOL DISTRICT
CLYMER CENTRAL SCHOOL DISTRICT
SCHOHARIE CENTRAL SCHOOL DISTRICT
HADLEY-LUZERNE CENTRAL SCHOOL DISTRICT
AVOCA CENTRAL SCHOOL DISTRICT
SCIO CENTRAL SCHOOL DISTRICT
MORRISVILLE-EATON CENTRAL SCHOOL DISTRICT
HAMILTON CENTRAL SCHOOL DISTRICT
STOCKBRIDGE VALLEY CENTRAL SCHOOL DISTRICT
MADISON CENTRAL SCHOOL DISTRICT
WORCESTER CENTRAL SCHOOL DISTRICT
HARRISVILLE CENTRAL SCHOOL DISTRICT
HANCOCK CENTRAL SCHOOL DISTRICT
FRANKLIN CENTRAL SCHOOL DISTRICT
ALEXANDRIA CENTRAL SCHOOL DISTRICT
SHARON SPRINGS CENTRAL SCHOOL DISTRICT
LYME CENTRAL SCHOOL DISTRICT
NEW LEBANON CENTRAL SCHOOL DISTRICT
HUNTER-TANNERSVILLE CENTRAL SCHOOL DISTRICT
ELBA CENTRAL SCHOOL DISTRICT
PAVILION CENTRAL SCHOOL DISTRICT
MORRIS CENTRAL SCHOOL DISTRICT
FORT ANN CENTRAL SCHOOL DISTRICT
NORTH WARREN CENTRAL SCHOOL DISTRICT
PANAMA CENTRAL SCHOOL DISTRICT
FORESTVILLE CENTRAL SCHOOL DISTRICT
HARTFORD CENTRAL SCHOOL DISTRICT
NEW YORK MILLS UNION FREE SCHOOL DISTRICT
PERRY CENTRAL SCHOOL DISTRICT
BOLIVAR-RICHBURG CENTRAL SCHOOL DISTRICT
CHATEAUGAY CENTRAL SCHOOL DISTRICT
GREEN ISLAND UNION FREE SCHOOL DISTRICT
HINSDALE CENTRAL SCHOOL DISTRICT
COPENHAGEN CENTRAL SCHOOL DISTRICT
MORIAH CENTRAL SCHOOL DISTRICT
CANAJOHARIE CENTRAL SCHOOL DISTRICT
EAST BLOOMFIELD CENTRAL SCHOOL DISTRICT
LAFAYETTE CENTRAL SCHOOL DISTRICT
RICHFIELD SPRINGS CENTRAL SCHOOL DISTRICT
EDMESTON CENTRAL SCHOOL DISTRICT
TIOGA CENTRAL SCHOOL DISTRICT
GROTON CENTRAL SCHOOL DISTRICT
NORTH COLLINS CENTRAL SCHOOL DISTRICT
ELDRED CENTRAL SCHOOL DISTRICT
NORTHERN ADIRONDACK CENTRAL SCHOOL DISTRICT
GEORGETOWN-SOUTH OTSELIC CENTRAL SCHOOL DISTRICT
PORTVILLE CENTRAL SCHOOL DISTRICT
ADDISON CENTRAL SCHOOL DISTRICT
ALFRED-ALMOND CENTRAL SCHOOL DISTRICT
DUNDEE CENTRAL SCHOOL DISTRICT
RED CREEK CENTRAL SCHOOL DISTRICT
ELLICOTTVILLE CENTRAL SCHOOL DISTRICT
FRANKLINVILLE CENTRAL SCHOOL DISTRICT
SALEM CENTRAL SCHOOL DISTRICT
ARGYLE CENTRAL SCHOOL DISTRICT
OPPENHEIM-EPHRATAH-ST. JOHNSVILLE CSD
WESTMORELAND CENTRAL SCHOOL DISTRICT
GREENWICH CENTRAL SCHOOL DISTRICT
SHERBURNE-EARLVILLE CENTRAL SCHOOL DISTRICT
PEMBROKE CENTRAL SCHOOL DISTRICT
WINDHAM-ASHLAND-JEWETT CENTRAL SCHOOL DISTRICT
LIVINGSTON MANOR CENTRAL SCHOOL DISTRICT
SPENCER-VAN ETTEN CENTRAL SCHOOL DISTRICT
KENDALL CENTRAL SCHOOL DISTRICT
PULASKI CENTRAL SCHOOL DISTRICT
TRUMANSBURG CENTRAL SCHOOL DISTRICT
DERUYTER CENTRAL SCHOOL DISTRICT
PINE VALLEY CENTRAL SCHOOL DISTRICT (SOUTH DAYTON)
WARRENSBURG CENTRAL SCHOOL DISTRICT
ADIRONDACK CENTRAL SCHOOL DISTRICT
HEUVELTON CENTRAL SCHOOL DISTRICT
MAYFIELD CENTRAL SCHOOL DISTRICT
CLIFTON-FINE CENTRAL SCHOOL DISTRICT
MIDDLEBURGH CENTRAL SCHOOL DISTRICT
HAMMONDSPORT CENTRAL SCHOOL DISTRICT
HERMON-DEKALB CENTRAL SCHOOL DISTRICT
LITTLE FALLS CITY SCHOOL DISTRICT
LAKE PLACID CENTRAL SCHOOL DISTRICT
SARANAC CENTRAL SCHOOL DISTRICT
AFTON CENTRAL SCHOOL DISTRICT
MARATHON CENTRAL SCHOOL DISTRICT
DELAWARE ACADEMY CENTRAL SCHOOL DISTRICT AT DELHI
BEAVER RIVER CENTRAL SCHOOL DISTRICT
OAKFIELD-ALABAMA CENTRAL SCHOOL DISTRICT
FRANKFORT-SCHUYLER CENTRAL SCHOOL DISTRICT
BAINBRIDGE-GUILFORD CENTRAL SCHOOL DISTRICT
HOLLAND CENTRAL SCHOOL DISTRICT
FORT EDWARD UNION FREE SCHOOL DISTRICT
BELLEVILLE-HENDERSON CENTRAL SCHOOL DISTRICT
MOUNT MARKHAM CENTRAL SCHOOL DISTRICT
CAMBRIDGE CENTRAL SCHOOL DISTRICT
WATERVILLE CENTRAL SCHOOL DISTRICT
SANDY CREEK CENTRAL SCHOOL DISTRICT
COOPERSTOWN CENTRAL SCHOOL DISTRICT
TICONDEROGA CENTRAL SCHOOL DISTRICT
EDEN CENTRAL SCHOOL DISTRICT
PINE PLAINS CENTRAL SCHOOL DISTRICT
LA FARGEVILLE CENTRAL SCHOOL DISTRICT
GREENVILLE CENTRAL SCHOOL DISTRICT
GERMANTOWN CENTRAL SCHOOL DISTRICT
WALTON CENTRAL SCHOOL DISTRICT
SOUTH KORTRIGHT CENTRAL SCHOOL DISTRICT
BERLIN CENTRAL SCHOOL DISTRICT
BRUNSWICK CENTRAL SCHOOL DISTRICT (BRITTONKILL)
DEPOSIT CENTRAL SCHOOL DISTRICT
HARPURSVILLE CENTRAL SCHOOL DISTRICT
BARKER CENTRAL SCHOOL DISTRICT
FALCONER CENTRAL SCHOOL DISTRICT
AKRON CENTRAL SCHOOL DISTRICT
MADRID-WADDINGTON CENTRAL SCHOOL DISTRICT
ODESSA-MONTOUR CENTRAL SCHOOL DISTRICT
CAMPBELL-SAVONA CENTRAL SCHOOL DISTRICT
ALEXANDER CENTRAL SCHOOL DISTRICT
DUANESBURG CENTRAL SCHOOL DISTRICT
ATTICA CENTRAL SCHOOL DISTRICT
LETCHWORTH CENTRAL SCHOOL DISTRICT
WELLSVILLE CENTRAL SCHOOL DISTRICT
RANDOLPH CENTRAL SCHOOL DISTRICT
SOUTH SENECA CENTRAL SCHOOL DISTRICT
AVON CENTRAL SCHOOL DISTRICT
EAST AURORA UNION FREE SCHOOL DISTRICT
ROMULUS CENTRAL SCHOOL DISTRICT
GALWAY CENTRAL SCHOOL DISTRICT
NEWARK VALLEY CENTRAL SCHOOL DISTRICT
SHELTER ISLAND UNION FREE SCHOOL DISTRICT
ORISKANY CENTRAL SCHOOL DISTRICT
TULLY CENTRAL SCHOOL DISTRICT
SKANEATELES CENTRAL SCHOOL DISTRICT
HOOSIC VALLEY CENTRAL SCHOOL DISTRICT
TUXEDO UNION FREE SCHOOL DISTRICT
DALTON-NUNDA CENTRAL SCHOOL DISTRICT (KESHEQUA)


CALEDONIA-MUMFORD CENTRAL SCHOOL DISTRICT
AUSABLE VALLEY CENTRAL SCHOOL DISTRICT
BROCTON CENTRAL SCHOOL DISTRICT
BEMUS POINT CENTRAL SCHOOL DISTRICT
SAUQUOIT VALLEY CENTRAL SCHOOL DISTRICT
SCHENEVUS CENTRAL SCHOOL DISTRICT
HANNIBAL CENTRAL SCHOOL DISTRICT
CHENANGO FORKS CENTRAL SCHOOL DISTRICT
SACKETS HARBOR CENTRAL SCHOOL DISTRICT
HOLLAND PATENT CENTRAL SCHOOL DISTRICT
GRANVILLE CENTRAL SCHOOL DISTRICT
FLORIDA UNION FREE SCHOOL DISTRICT
UNION SPRINGS CENTRAL SCHOOL DISTRICT
CANTON CENTRAL SCHOOL DISTRICT
CANISTEO-GREENWOOD CSD
WAYLAND-COHOCTON CENTRAL SCHOOL DISTRICT
MILLBROOK CENTRAL SCHOOL DISTRICT
NEWFIELD CENTRAL SCHOOL DISTRICT
WAVERLY CENTRAL SCHOOL DISTRICT
WEST CANADA VALLEY CENTRAL SCHOOL DISTRICT
NAPLES CENTRAL SCHOOL DISTRICT
OXFORD ACADEMY AND CENTRAL SCHOOL DISTRICT
GREENE CENTRAL SCHOOL DISTRICT
POLAND CENTRAL SCHOOL DISTRICT
SCHODACK CENTRAL SCHOOL DISTRICT
MARCELLUS CENTRAL SCHOOL DISTRICT
GORHAM-MIDDLESEX CENTRAL SCHOOL DISTRICT (MARCUS WHITMAN)
BATH CENTRAL SCHOOL DISTRICT
CATO-MERIDIAN CENTRAL SCHOOL DISTRICT
MARION CENTRAL SCHOOL DISTRICT
GANANDA CENTRAL SCHOOL DISTRICT
EAST WILLISTON UNION FREE SCHOOL DISTRICT
WHITNEY POINT CENTRAL SCHOOL DISTRICT
MORAVIA CENTRAL SCHOOL DISTRICT
SOUTHWESTERN CENTRAL SCHOOL DISTRICT AT JAMESTOWN
ALTMAR-PARISH-WILLIAMSTOWN CENTRAL SCHOOL DISTRICT
CHAUTAUQUA LAKE CENTRAL SCHOOL DISTRICT
MILFORD CENTRAL SCHOOL DISTRICT
CASSADAGA VALLEY CENTRAL SCHOOL DISTRICT
BRUSHTON-MOIRA CENTRAL SCHOOL DISTRICT
BRIARCLIFF MANOR UNION FREE SCHOOL DISTRICT
SCHUYLERVILLE CENTRAL SCHOOL DISTRICT
WATERFORD-HALFMOON UNION FREE SCHOOL DISTRICT
NORTH SALEM CENTRAL SCHOOL DISTRICT
BYRAM HILLS CENTRAL SCHOOL DISTRICT
PORT BYRON CENTRAL SCHOOL DISTRICT
EDWARDS-KNOX CENTRAL SCHOOL DISTRICT
CLINTON CENTRAL SCHOOL DISTRICT
BRONXVILLE UNION FREE SCHOOL DISTRICT
CANDOR CENTRAL SCHOOL DISTRICT
LE ROY CENTRAL SCHOOL DISTRICT
SOUTH LEWIS CENTRAL SCHOOL DISTRICT
SULLIVAN WEST CENTRAL SCHOOL DISTRICT
ELMIRA HEIGHTS CENTRAL SCHOOL DISTRICT
BRIDGEHAMPTON UNION FREE SCHOOL DISTRICT
BEEKMANTOWN CENTRAL SCHOOL DISTRICT
MARGARETVILLE CENTRAL SCHOOL DISTRICT
HERKIMER CENTRAL SCHOOL DISTRICT
LIVONIA CENTRAL SCHOOL DISTRICT
EDGEMONT UNION FREE SCHOOL DISTRICT
VOORHEESVILLE CENTRAL SCHOOL DISTRICT
RHINEBECK CENTRAL SCHOOL DISTRICT
NORTHEAST CENTRAL SCHOOL DISTRICT
CHEEKTOWAGA-SLOAN UNION FREE SCHOOL DISTRICT
ALDEN CENTRAL SCHOOL DISTRICT
CORINTH CENTRAL SCHOOL DISTRICT
POTSDAM CENTRAL SCHOOL DISTRICT
SOUTHERN CAYUGA CENTRAL SCHOOL DISTRICT
TUCKAHOE UNION FREE SCHOOL DISTRICT
SHERRILL CITY SCHOOL DISTRICT
YORK CENTRAL SCHOOL DISTRICT
LAKE GEORGE CENTRAL SCHOOL DISTRICT
LYNDONVILLE CENTRAL SCHOOL DISTRICT
WHITEHALL CENTRAL SCHOOL DISTRICT
THOUSAND ISLANDS CENTRAL SCHOOL DISTRICT
ALLEGANY-LIMESTONE CENTRAL SCHOOL DISTRICT
CATTARAUGUS-LITTLE VALLEY CENTRAL SCHOOL DISTRICT
FORT PLAIN CENTRAL SCHOOL DISTRICT
CUBA-RUSHFORD CENTRAL SCHOOL DISTRICT
BLIND BROOK-RYE UNION FREE SCHOOL DISTRICT
GENERAL BROWN CENTRAL SCHOOL DISTRICT
HOLLEY CENTRAL SCHOOL DISTRICT
ONEONTA CITY SCHOOL DISTRICT
OTEGO-UNADILLA CENTRAL SCHOOL DISTRICT
NORWOOD-NORFOLK CENTRAL SCHOOL DISTRICT
SALMON RIVER CENTRAL SCHOOL DISTRICT
BROADALBIN-PERTH CENTRAL SCHOOL DISTRICT
WESTFIELD CENTRAL SCHOOL DISTRICT
SILVER CREEK CENTRAL SCHOOL DISTRICT
OWEGO-APALACHIN CENTRAL SCHOOL DISTRICT
SHOREHAM-WADING RIVER CENTRAL SCHOOL DISTRICT
SAG HARBOR UNION FREE SCHOOL DISTRICT
MILLER PLACE UNION FREE SCHOOL DISTRICT
FABIUS-POMPEY CENTRAL SCHOOL DISTRICT
CANASTOTA CENTRAL SCHOOL DISTRICT
WAYNE CENTRAL SCHOOL DISTRICT
WANTAGH UNION FREE SCHOOL DISTRICT
SEAFORD UNION FREE SCHOOL DISTRICT
WILSON CENTRAL SCHOOL DISTRICT
CROTON-HARMON UNION FREE SCHOOL DISTRICT
STILLWATER CENTRAL SCHOOL DISTRICT
WATKINS GLEN CENTRAL SCHOOL DISTRICT
HORNELL CITY SCHOOL DISTRICT
BYRON-BERGEN CENTRAL SCHOOL DISTRICT
SCOTIA-GLENVILLE CENTRAL SCHOOL DISTRICT
SARANAC LAKE CENTRAL SCHOOL DISTRICT
HOOSICK FALLS CENTRAL SCHOOL DISTRICT
HALDANE CENTRAL SCHOOL DISTRICT
CENTRAL VALLEY CSD AT ILION-MOHAWK
CHESTER UNION FREE SCHOOL DISTRICT
LEWISTON-PORTER CENTRAL SCHOOL DISTRICT
MINEOLA UNION FREE SCHOOL DISTRICT
GOWANDA CENTRAL SCHOOL DISTRICT
WHEATLAND-CHILI CENTRAL SCHOOL DISTRICT
HOMER CENTRAL SCHOOL DISTRICT
CHITTENANGO CENTRAL SCHOOL DISTRICT
CLYDE-SAVANNAH CENTRAL SCHOOL DISTRICT
ONEIDA CITY SCHOOL DISTRICT
WEEDSPORT CENTRAL SCHOOL DISTRICT
KATONAH-LEWISBORO UNION FREE SCHOOL DISTRICT
FREWSBURG CENTRAL SCHOOL DISTRICT
MANCHESTER-SHORTSVILLE CENTRAL SCHOOL DISTRICT (RED JACKET)
DRYDEN CENTRAL SCHOOL DISTRICT
COLD SPRING HARBOR CENTRAL SCHOOL DISTRICT
HASTINGS-ON-HUDSON UNION FREE SCHOOL DISTRICT
SUSQUEHANNA VALLEY CENTRAL SCHOOL DISTRICT
PLEASANTVILLE UNION FREE SCHOOL DISTRICT
HAMBURG CENTRAL SCHOOL DISTRICT
TONAWANDA CITY SCHOOL DISTRICT
OGDENSBURG CITY SCHOOL DISTRICT
PEARL RIVER UNION FREE SCHOOL DISTRICT
SPRINGVILLE-GRIFFITH INSTITUTE CENTRAL SCHOOL DISTRICT
CAIRO-DURHAM CENTRAL SCHOOL DISTRICT
COXSACKIE-ATHENS CENTRAL SCHOOL DISTRICT
WINDSOR CENTRAL SCHOOL DISTRICT
PHOENIX CENTRAL SCHOOL DISTRICT
SIDNEY CENTRAL SCHOOL DISTRICT
CENTER MORICHES UNION FREE SCHOOL DISTRICT
SOUTHOLD UNION FREE SCHOOL DISTRICT
LOWVILLE ACADEMY & CENTRAL SCHOOL DISTRICT
RYE CITY SCHOOL DISTRICT
MOUNT PLEASANT CENTRAL SCHOOL DISTRICT
GARDEN CITY UNION FREE SCHOOL DISTRICT
EAST ROCHESTER UNION FREE SCHOOL DISTRICT
JORDAN-ELBRIDGE CENTRAL SCHOOL DISTRICT
IRVINGTON UNION FREE SCHOOL DISTRICT
DOBBS FERRY UNION FREE SCHOOL DISTRICT
COHOES CITY SCHOOL DISTRICT
WARSAW CENTRAL SCHOOL DISTRICT
SCARSDALE UNION FREE SCHOOL DISTRICT
SOMERS CENTRAL SCHOOL DISTRICT
LYONS CENTRAL SCHOOL DISTRICT
WILLIAMSON CENTRAL SCHOOL DISTRICT
TRI-VALLEY CENTRAL SCHOOL DISTRICT
ROOSEVELT UNION FREE SCHOOL DISTRICT
MOUNT MORRIS CENTRAL SCHOOL DISTRICT
WESTHILL CENTRAL SCHOOL DISTRICT
NEWFANE CENTRAL SCHOOL DISTRICT
CAMDEN CENTRAL SCHOOL DISTRICT
CAZENOVIA CENTRAL SCHOOL DISTRICT
WHITESBORO CENTRAL SCHOOL DISTRICT
STARPOINT CENTRAL SCHOOL DISTRICT
WEST ISLIP UNION FREE SCHOOL DISTRICT
PERU CENTRAL SCHOOL DISTRICT
SPACKENKILL UNION FREE SCHOOL DISTRICT
PHELPS-CLIFTON SPRINGS CENTRAL SCHOOL DISTRICT
MECHANICVILLE CITY SCHOOL DISTRICT
GLOVERSVILLE CITY SCHOOL DISTRICT
AVERILL PARK CENTRAL SCHOOL DISTRICT
BURNT HILLS-BALLSTON LAKE CENTRAL SCHOOL DISTRICT
WATERLOO CENTRAL SCHOOL DISTRICT
CENTRAL SQUARE CENTRAL SCHOOL DISTRICT
SOUTH GLENS FALLS CENTRAL SCHOOL DISTRICT
CHENANGO VALLEY CENTRAL SCHOOL DISTRICT
CHATHAM CENTRAL SCHOOL DISTRICT
CORTLAND CITY SCHOOL DISTRICT
SAYVILLE UNION FREE SCHOOL DISTRICT
NEW HARTFORD CENTRAL SCHOOL DISTRICT
DOVER UNION FREE SCHOOL DISTRICT
SALAMANCA CITY SCHOOL DISTRICT
DANSVILLE CENTRAL SCHOOL DISTRICT
CLARENCE CENTRAL SCHOOL DISTRICT
PORT JEFFERSON UNION FREE SCHOOL DISTRICT
FONDA-FULTONVILLE CENTRAL SCHOOL DISTRICT
BRASHER FALLS CENTRAL SCHOOL DISTRICT
MEXICO CENTRAL SCHOOL DISTRICT
BABYLON UNION FREE SCHOOL DISTRICT
SENECA FALLS CENTRAL SCHOOL DISTRICT
HAMPTON BAYS UNION FREE SCHOOL DISTRICT
GREENPORT UNION FREE SCHOOL DISTRICT
UNADILLA VALLEY CENTRAL SCHOOL DISTRICT
NORWICH CITY SCHOOL DISTRICT
NORTH SHORE CENTRAL SCHOOL DISTRICT
EAST ROCKAWAY UNION FREE SCHOOL DISTRICT
CARLE PLACE UNION FREE SCHOOL DISTRICT
MOUNT SINAI UNION FREE SCHOOL DISTRICT
ALBION CENTRAL SCHOOL DISTRICT
COBLESKILL-RICHMONDVILLE CENTRAL SCHOOL DISTRICT
CHAPPAQUA CENTRAL SCHOOL DISTRICT
ARDSLEY UNION FREE SCHOOL DISTRICT
School District


NORTH ROSE-WOLCOTT CENTRAL SCHOOL DISTRICT
BETHPAGE UNION FREE SCHOOL DISTRICT
YORKSHIRE-PIONEER CENTRAL SCHOOL DISTRICT
ROYALTON-HARTLAND CENTRAL SCHOOL DISTRICT
ONONDAGA CENTRAL SCHOOL DISTRICT
EVANS-BRANT CENTRAL SCHOOL DISTRICT (LAKE SHORE)
CLEVELAND HILL UNION FREE SCHOOL DISTRICT
HUDSON FALLS CENTRAL SCHOOL DISTRICT
HONEOYE FALLS-LIMA CENTRAL SCHOOL DISTRICT
NORTHEASTERN CLINTON CENTRAL SCHOOL DISTRICT
PITTSFORD CENTRAL SCHOOL DISTRICT
PLAINEDGE UNION FREE SCHOOL DISTRICT
ISLAND TREES UNION FREE SCHOOL DISTRICT
RAVENA-COEYMANS-SELKIRK CENTRAL SCHOOL DISTRICT
RYE NECK UNION FREE SCHOOL DISTRICT
PELHAM UNION FREE SCHOOL DISTRICT
EASTPORT-SOUTH MANOR CSD
MARLBORO CENTRAL SCHOOL DISTRICT
KINGS PARK CENTRAL SCHOOL DISTRICT
GLENS FALLS CITY SCHOOL DISTRICT
JERICHO UNION FREE SCHOOL DISTRICT
BATAVIA CITY SCHOOL DISTRICT
NANUET UNION FREE SCHOOL DISTRICT
ONTEORA CENTRAL SCHOOL DISTRICT
HAUPPAUGE UNION FREE SCHOOL DISTRICT
EAST HAMPTON UNION FREE SCHOOL DISTRICT
ELLENVILLE CENTRAL SCHOOL DISTRICT
LANSING CENTRAL SCHOOL DISTRICT
MINISINK VALLEY CENTRAL SCHOOL DISTRICT
IROQUOIS CENTRAL SCHOOL DISTRICT
RED HOOK CENTRAL SCHOOL DISTRICT
DEPEW UNION FREE SCHOOL DISTRICT
CHEEKTOWAGA-MARYVALE UNION FREE SCHOOL DISTRICT
SOLVAY UNION FREE SCHOOL DISTRICT
WYANDANCH UNION FREE SCHOOL DISTRICT
GENESEO CENTRAL SCHOOL DISTRICT
MASSAPEQUA UNION FREE SCHOOL DISTRICT
MATTITUCK-CUTCHOGUE UNION FREE SCHOOL DISTRICT
LAWRENCE UNION FREE SCHOOL DISTRICT
FREDONIA CENTRAL SCHOOL DISTRICT
EASTCHESTER UNION FREE SCHOOL DISTRICT
OLEAN CITY SCHOOL DISTRICT
OYSTER BAY-EAST NORWICH CENTRAL SCHOOL DISTRICT
PALMYRA-MACEDON CENTRAL SCHOOL DISTRICT
LOCUST VALLEY CENTRAL SCHOOL DISTRICT
BETHLEHEM CENTRAL SCHOOL DISTRICT
RONDOUT VALLEY CENTRAL SCHOOL DISTRICT
PUTNAM VALLEY CENTRAL SCHOOL DISTRICT
ELMSFORD UNION FREE SCHOOL DISTRICT
UNION-ENDICOTT CENTRAL SCHOOL DISTRICT
YORKTOWN CENTRAL SCHOOL DISTRICT
BALLSTON SPA CENTRAL SCHOOL DISTRICT
MEDINA CENTRAL SCHOOL DISTRICT
LANCASTER CENTRAL SCHOOL DISTRICT
JOHNSTOWN CITY SCHOOL DISTRICT
GOUVERNEUR CENTRAL SCHOOL DISTRICT
RENSSELAER CITY SCHOOL DISTRICT
SCHALMONT CENTRAL SCHOOL DISTRICT
CHURCHVILLE-CHILI CENTRAL SCHOOL DISTRICT
ELWOOD UNION FREE SCHOOL DISTRICT
PLAINVIEW-OLD BETHPAGE CENTRAL SCHOOL DISTRICT
ROCKVILLE CENTRE UNION FREE SCHOOL DISTRICT
MANHASSET UNION FREE SCHOOL DISTRICT
PINE BUSH CENTRAL SCHOOL DISTRICT
PAWLING CENTRAL SCHOOL DISTRICT
FRONTIER CENTRAL SCHOOL DISTRICT
HENDRICK HUDSON CENTRAL SCHOOL DISTRICT
PENN YAN CENTRAL SCHOOL DISTRICT
BAYPORT-BLUE POINT UNION FREE SCHOOL DISTRICT
BROCKPORT CENTRAL SCHOOL DISTRICT
LONG BEACH CITY SCHOOL DISTRICT
PORT CHESTER-RYE UNION FREE SCHOOL DISTRICT
SOUTHAMPTON UNION FREE SCHOOL DISTRICT
ROSLYN UNION FREE SCHOOL DISTRICT
ELMIRA CITY SCHOOL DISTRICT
EAST MEADOW UNION FREE SCHOOL DISTRICT
VALHALLA UNION FREE SCHOOL DISTRICT
MAINE-ENDWELL CENTRAL SCHOOL DISTRICT
MASSENA CENTRAL SCHOOL DISTRICT
MALONE CENTRAL SCHOOL DISTRICT
ISLIP UNION FREE SCHOOL DISTRICT
HILTON CENTRAL SCHOOL DISTRICT
COMMACK UNION FREE SCHOOL DISTRICT
NORTH TONAWANDA CITY SCHOOL DISTRICT
EAST SYRACUSE MINOA CENTRAL SCHOOL DISTRICT
ROCKY POINT UNION FREE SCHOOL DISTRICT
CANANDAIGUA CITY SCHOOL DISTRICT
BROOKHAVEN-COMSEWOGUE UNION FREE SCHOOL DISTRICT
VICTOR CENTRAL SCHOOL DISTRICT
CATSKILL CENTRAL SCHOOL DISTRICT
LANSINGBURGH CENTRAL SCHOOL DISTRICT
EAST GREENBUSH CENTRAL SCHOOL DISTRICT
HUNTINGTON UNION FREE SCHOOL DISTRICT
HARBORFIELDS CENTRAL SCHOOL DISTRICT
HERRICKS UNION FREE SCHOOL DISTRICT
PEEKSKILL CITY SCHOOL DISTRICT
THREE VILLAGE CENTRAL SCHOOL DISTRICT
TACONIC HILLS CENTRAL SCHOOL DISTRICT
EAST ISLIP UNION FREE SCHOOL DISTRICT
BEACON CITY SCHOOL DISTRICT
SODUS CENTRAL SCHOOL DISTRICT
NEWARK CENTRAL SCHOOL DISTRICT
HICKSVILLE UNION FREE SCHOOL DISTRICT
SYOSSET CENTRAL SCHOOL DISTRICT
HIGHLAND CENTRAL SCHOOL DISTRICT
UNION FREE SCHOOL DISTRICT OF THE TARRYTOWNS
HORSEHEADS CENTRAL SCHOOL DISTRICT
GLEN COVE CITY SCHOOL DISTRICT
WARWICK VALLEY CENTRAL SCHOOL DISTRICT
SWEET HOME CENTRAL SCHOOL DISTRICT
WEST GENESEE CENTRAL SCHOOL DISTRICT
CARTHAGE CENTRAL SCHOOL DISTRICT
ROTTERDAM-MOHONASEN CENTRAL SCHOOL DISTRICT
WEST BABYLON UNION FREE SCHOOL DISTRICT
CARMEL CENTRAL SCHOOL DISTRICT
MAHOPAC CENTRAL SCHOOL DISTRICT
NYACK UNION FREE SCHOOL DISTRICT
SUFFERN CENTRAL SCHOOL DISTRICT
THE ENLARGED CITY SCHOOL DISTRICT OF THE CITY OF SARATOGA SP
LAKELAND CENTRAL SCHOOL DISTRICT
GREENBURGH CENTRAL SCHOOL DISTRICT
HARRISON CENTRAL SCHOOL DISTRICT
OSSINING UNION FREE SCHOOL DISTRICT
MAMARONECK UNION FREE SCHOOL DISTRICT
JOHNSON CITY CENTRAL SCHOOL DISTRICT
SOUTH ORANGETOWN CENTRAL SCHOOL DISTRICT
WHITE PLAINS CITY SCHOOL DISTRICT
OSWEGO CITY SCHOOL DISTRICT
BREWSTER CENTRAL SCHOOL DISTRICT
SOUTH JEFFERSON CENTRAL SCHOOL DISTRICT
NISKAYUNA CENTRAL SCHOOL DISTRICT
DEER PARK UNION FREE SCHOOL DISTRICT
NIAGARA-WHEATFIELD CENTRAL SCHOOL DISTRICT
SOUTH HUNTINGTON UNION FREE SCHOOL DISTRICT
EAST IRONDEQUOIT CENTRAL SCHOOL DISTRICT
WEST IRONDEQUOIT CENTRAL SCHOOL DISTRICT
JAMESVILLE-DEWITT CENTRAL SCHOOL DISTRICT
AMHERST CENTRAL SCHOOL DISTRICT
LINDENHURST UNION FREE SCHOOL DISTRICT
KINDERHOOK CENTRAL SCHOOL DISTRICT
PLATTSBURGH CITY SCHOOL DISTRICT
MALVERNE UNION FREE SCHOOL DISTRICT
OCEANSIDE UNION FREE SCHOOL DISTRICT
LIBERTY CENTRAL SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #32
FAIRPORT CENTRAL SCHOOL DISTRICT
CONNETQUOT CENTRAL SCHOOL DISTRICT
NORTHPORT-EAST NORTHPORT UNION FREE SCHOOL DISTRICT
BEDFORD CENTRAL SCHOOL DISTRICT
FARMINGDALE UNION FREE SCHOOL DISTRICT
GUILDERLAND CENTRAL SCHOOL DISTRICT
PORT WASHINGTON UNION FREE SCHOOL DISTRICT
WEST HEMPSTEAD UNION FREE SCHOOL DISTRICT
WALLKILL CENTRAL SCHOOL DISTRICT
GREAT NECK UNION FREE SCHOOL DISTRICT
LYNBROOK UNION FREE SCHOOL DISTRICT
WASHINGTONVILLE CENTRAL SCHOOL DISTRICT
FAYETTEVILLE-MANLIUS CENTRAL SCHOOL DISTRICT
BAY SHORE UNION FREE SCHOOL DISTRICT
COPIAGUE UNION FREE SCHOOL DISTRICT
UTICA CITY SCHOOL DISTRICT
SPENCERPORT CENTRAL SCHOOL DISTRICT
HEWLETT-WOODMERE UNION FREE SCHOOL DISTRICT
CORNING CITY SCHOOL DISTRICT
HUDSON CITY SCHOOL DISTRICT
MONROE-WOODBURY CENTRAL SCHOOL DISTRICT
GOSHEN CENTRAL SCHOOL DISTRICT
PORT JERVIS CITY SCHOOL DISTRICT
GRAND ISLAND CENTRAL SCHOOL DISTRICT
VESTAL CENTRAL SCHOOL DISTRICT
SHENENDEHOWA CENTRAL SCHOOL DISTRICT
WESTHAMPTON BEACH UNION FREE SCHOOL DISTRICT
ORCHARD PARK CENTRAL SCHOOL DISTRICT
WESTBURY UNION FREE SCHOOL DISTRICT
SAUGERTIES CENTRAL SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT # 8
NEW YORK CITY GEOGRAPHIC DISTRICT # 9
LACKAWANNA CITY SCHOOL DISTRICT
EAST RAMAPO CENTRAL SCHOOL DISTRICT (SPRING VALLEY)
MIDDLE COUNTRY CENTRAL SCHOOL DISTRICT
LIVERPOOL CENTRAL SCHOOL DISTRICT
CORNWALL CENTRAL SCHOOL DISTRICT
GENEVA CITY SCHOOL DISTRICT
HIGHLAND FALLS CENTRAL SCHOOL DISTRICT
CHEEKTOWAGA CENTRAL SCHOOL DISTRICT
INDIAN RIVER CENTRAL SCHOOL DISTRICT
FULTON CITY SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #21
NEW YORK CITY GEOGRAPHIC DISTRICT #22
NEW YORK CITY GEOGRAPHIC DISTRICT #23
NEW YORK CITY GEOGRAPHIC DISTRICT #28
NEW YORK CITY GEOGRAPHIC DISTRICT #27
NEW YORK CITY GEOGRAPHIC DISTRICT #24
NEW YORK CITY GEOGRAPHIC DISTRICT #25
NEW YORK CITY GEOGRAPHIC DISTRICT #30
LEVITTOWN UNION FREE SCHOOL DISTRICT
SMITHTOWN CENTRAL SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #29
UNIONDALE UNION FREE SCHOOL DISTRICT
PENFIELD CENTRAL SCHOOL DISTRICT


BRIGHTON CENTRAL SCHOOL DISTRICT
WAPPINGERS CENTRAL SCHOOL DISTRICT
NIAGARA FALLS CITY SCHOOL DISTRICT
HALF HOLLOW HILLS CENTRAL SCHOOL DISTRICT
BALDWIN UNION FREE SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT # 4
ITHACA CITY SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT # 1
AUBURN CITY SCHOOL DISTRICT
YONKERS CITY SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT # 5
NEW YORK CITY GEOGRAPHIC DISTRICT # 3
NEW YORK CITY GEOGRAPHIC DISTRICT # 7
NEW YORK CITY GEOGRAPHIC DISTRICT # 6
NEW ROCHELLE CITY SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #13
BINGHAMTON CITY SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #10
NEW YORK CITY GEOGRAPHIC DISTRICT #11
NEW YORK CITY GEOGRAPHIC DISTRICT #19
NEW PALTZ CENTRAL SCHOOL DISTRICT
MONTICELLO CENTRAL SCHOOL DISTRICT
NORTH COLONIE CSD
NEW YORK CITY GEOGRAPHIC DISTRICT #15
WATERVLIET CITY SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #16
NEW YORK CITY GEOGRAPHIC DISTRICT #18
NEW YORK CITY GEOGRAPHIC DISTRICT # 2
QUEENSBURY UNION FREE SCHOOL DISTRICT
LONGWOOD CENTRAL SCHOOL DISTRICT
LOCKPORT CITY SCHOOL DISTRICT
WEBSTER CENTRAL SCHOOL DISTRICT
AMSTERDAM CITY SCHOOL DISTRICT
CENTRAL ISLIP UNION FREE SCHOOL DISTRICT
WILLIAM FLOYD UNION FREE SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #20
HEMPSTEAD UNION FREE SCHOOL DISTRICT
FREEPORT UNION FREE SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #14
SACHEM CENTRAL SCHOOL DISTRICT
MOUNT VERNON SCHOOL DISTRICT
HAVERSTRAW-STONY POINT CSD (NORTH ROCKLAND)
CLARKSTOWN CENTRAL SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #12
NEW YORK CITY GEOGRAPHIC DISTRICT #26
NORTH BABYLON UNION FREE SCHOOL DISTRICT
POUGHKEEPSIE CITY SCHOOL DISTRICT
HYDE PARK CENTRAL SCHOOL DISTRICT
SCHENECTADY CITY SCHOOL DISTRICT
BALDWINSVILLE CENTRAL SCHOOL DISTRICT
AMITYVILLE UNION FREE SCHOOL DISTRICT
BELLMORE-MERRICK CENTRAL HIGH SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #17
SOUTH COLONIE CENTRAL SCHOOL DISTRICT
VALLEY STREAM CENTRAL HIGH SCHOOL DISTRICT
SOUTH COUNTRY CENTRAL SCHOOL DISTRICT
SEWANHAKA CENTRAL HIGH SCHOOL DISTRICT
WATERTOWN CITY SCHOOL DISTRICT
TROY CITY SCHOOL DISTRICT
ARLINGTON CENTRAL SCHOOL DISTRICT
NEW YORK CITY GEOGRAPHIC DISTRICT #31
FALLSBURG CENTRAL SCHOOL DISTRICT
NORTH SYRACUSE CENTRAL SCHOOL DISTRICT
JAMESTOWN CITY SCHOOL DISTRICT
DUNKIRK CITY SCHOOL DISTRICT
PATCHOGUE-MEDFORD UNION FREE SCHOOL DISTRICT
ROME CITY SCHOOL DISTRICT
SYRACUSE CITY SCHOOL DISTRICT
NEWBURGH CITY SCHOOL DISTRICT
BUFFALO CITY SCHOOL DISTRICT
WEST SENECA CENTRAL SCHOOL DISTRICT
GATES-CHILI CENTRAL SCHOOL DISTRICT
VALLEY CENTRAL SCHOOL DISTRICT (MONTGOMERY)
BRENTWOOD UNION FREE SCHOOL DISTRICT
RUSH-HENRIETTA CENTRAL SCHOOL DISTRICT
ALBANY CITY SCHOOL DISTRICT
RIVERHEAD CENTRAL SCHOOL DISTRICT
KENMORE-TONAWANDA UNION FREE SCHOOL DISTRICT
GREECE CENTRAL SCHOOL DISTRICT
ROCHESTER CITY SCHOOL DISTRICT
MIDDLETOWN CITY SCHOOL DISTRICT
WILLIAMSVILLE CENTRAL SCHOOL DISTRICT
KINGSTON CITY SCHOOL DISTRICT
0
20
40
60
80
100
120
140
Count

The bar chart further satisfies the claim of evenly spread observations for districts.

nrc_code

```
#nrc_code
df['nrc_code'].describe()
```

	nrc_code
count	73152.000000
mean	4.588583
std	1.203507
min	1.000000
25%	4.000000
50%	5.000000
75%	5.000000
max	6.000000

dtype: float64

```
print(df['nrc_code'].isnull().sum())
mapping = df['nrc_code'].unique()
print(mapping)
```

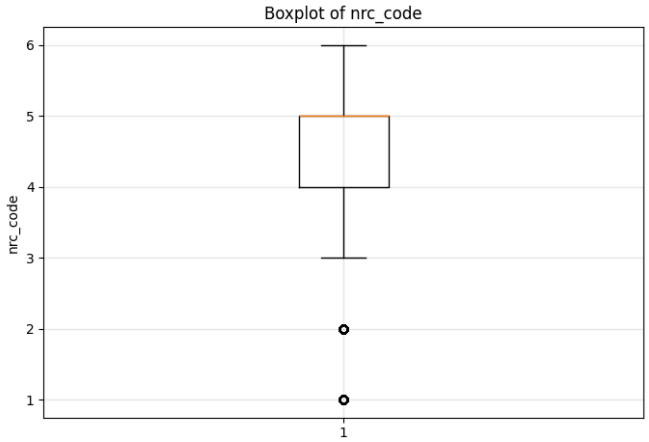
```
0
[3 5 6 4 2 1]
```

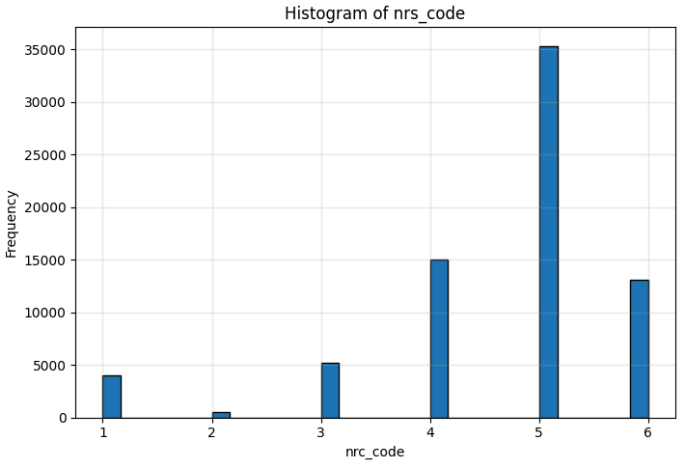
```
# Create side-by-side plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Boxplot
axes[0].boxplot(df['nrc_code'])
axes[0].set_ylabel('nrc_code')
axes[0].set_title('Boxplot of nrc_code')
axes[0].grid(True, alpha=0.3)

# Histogram
axes[1].hist(df['nrc_code'], bins=30, edgecolor='black')
axes[1].set_xlabel('nrc_code')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of nrs_code')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





The nrc code has values from 1 to 6 and is left skewed as per the histogram.

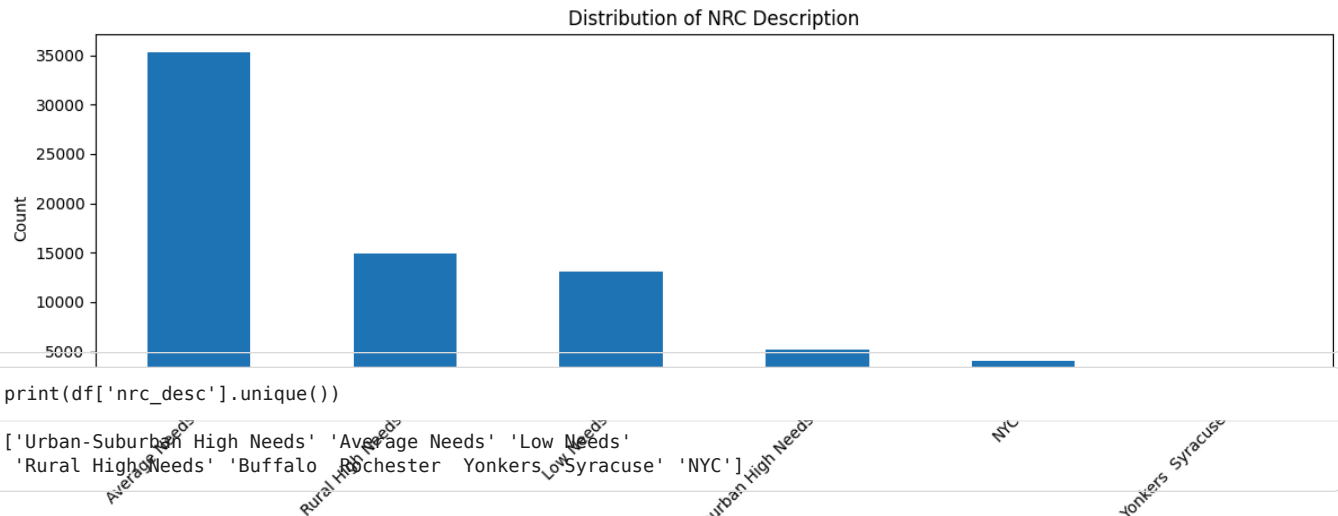
nrc_desc

```
#nrc_desc
df['nrc_desc'].describe()
```

	nrc_desc
count	73152
unique	6
top	Average Needs
freq	35322

dtype: object

```
 # Create bar chart for nrc_desc
nrc_desc_counts = df['nrc_desc'].value_counts()
plt.figure(figsize=(12, 6))
nrc_desc_counts.plot(kind='bar')
plt.xlabel('NRC Description')
plt.ylabel('Count')
plt.title('Distribution of NRC Description')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

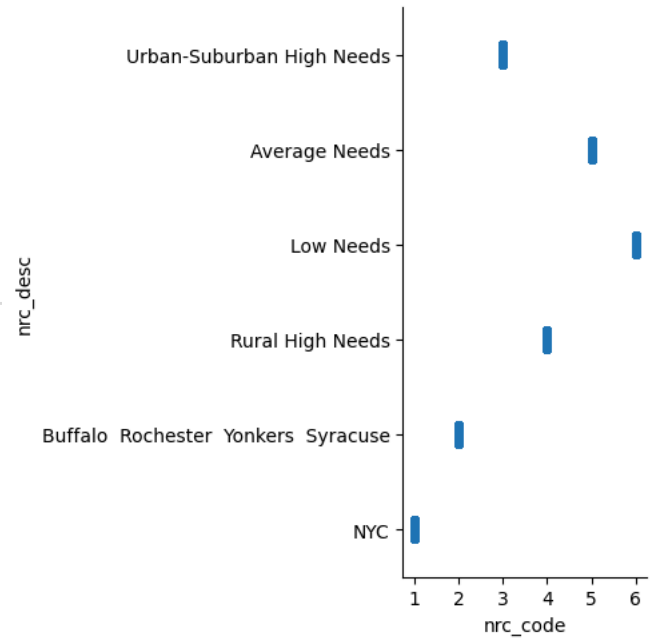



```
print(df['nrc_desc'].unique())
```

```
['Urban-Suburban High Needs' 'Average Needs' 'Low Needs'
 'Rural High Needs' 'Buffalo  Rochester  Yonkers  Syracuse' 'NYC']
```

nrc_code & nrc_desc

```
import seaborn as sns
sns.catplot(data=df, x="nrc_code", y="nrc_desc", kind='strip')
plt.xticks([1, 2, 3, 4, 5, 6])  # Force ticks at each integer
plt.show()
```



Here, the nrc_code and nrc_desc represent the same thing and categorical column is nominal. We will keep the nrc_desc column and drop the nrc_code and then perform one_hot encoding for use on the model.

county_code

```
#county_code
df['county_code'].describe()
```

	county_code
count	73152.000000
mean	36.251859
std	20.588044
min	1.000000
25%	17.000000
50%	40.000000
75%	57.000000
max	68.000000

dtype: float64

```
print(df['county_code'].nunique())
df['county_code'].unique()
```

```
62
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16, 17,
       18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 31, 32, 33, 34, 35, 40,
       41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57,
       58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68])
```

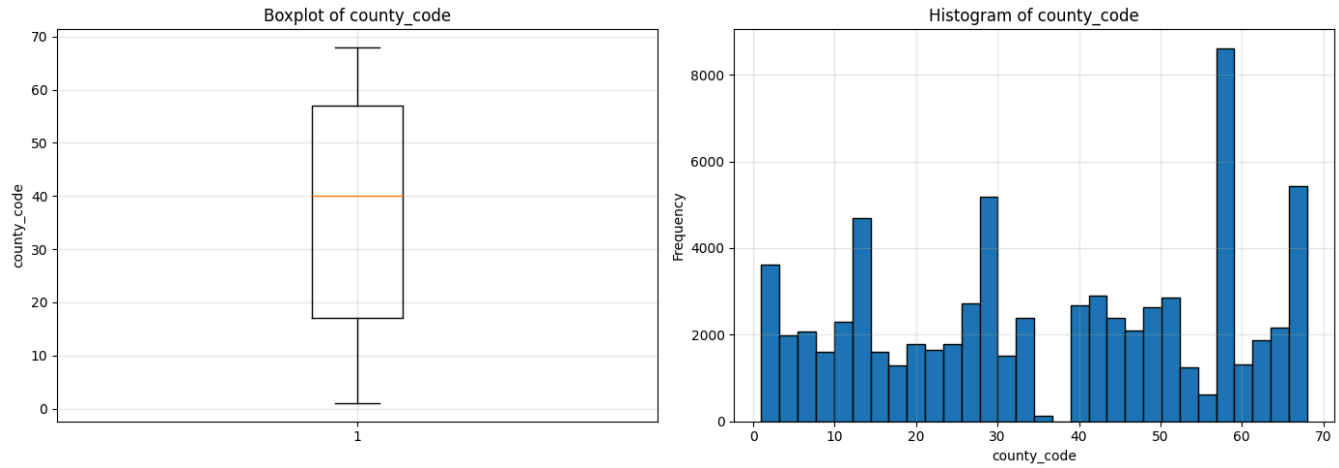
There are 62 unique county codes

```
# Create side-by-side plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Boxplot
axes[0].boxplot(df['county_code'])
axes[0].set_ylabel('county_code')
axes[0].set_title('Boxplot of county_code')
axes[0].grid(True, alpha=0.3)

# Histogram
axes[1].hist(df['county_code'], bins=30, edgecolor='black')
axes[1].set_xlabel('county_code')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of county_code')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



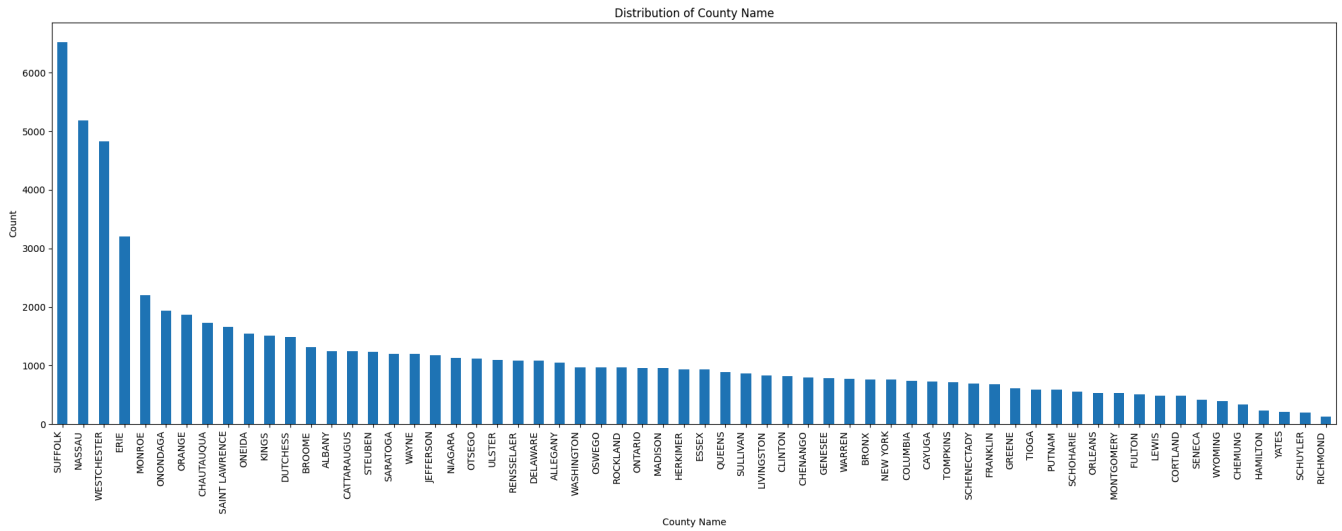
county_name

```
#county_name
df['county_name'].describe()
```

	county_name
count	73152
unique	62
top	SUFFOLK
freq	6526

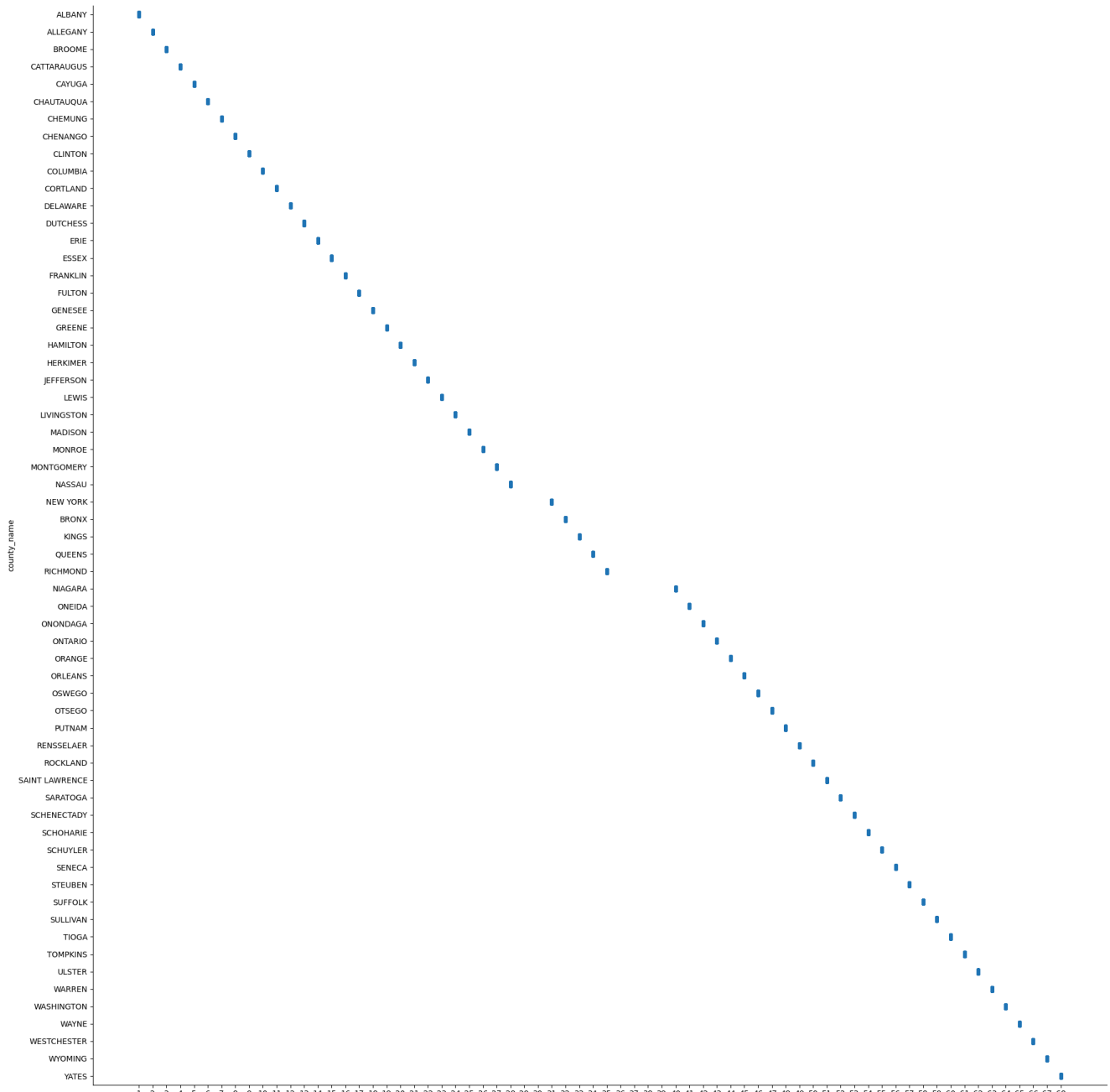
dtype: object

```
  # Create bar chart for county_name
county_counts = df['county_name'].value_counts()
plt.figure(figsize=(20, 8))
county_counts.plot(kind='bar')
plt.xlabel('County Name')
plt.ylabel('Count')
plt.title('Distribution of County Name')
plt.xticks(rotation=90, ha='right')
plt.tight_layout()
plt.show()
```



county_code & county_name

```
import numpy as np
g = sns.catplot(data=df, x="county_code", y="county_name", kind='strip', height=20)
g.ax.set_xticks(np.arange(df['county_code'].min(), df['county_code'].max() + 1, 1))
plt.show()
```


county_name
ALBANY
ALLEGANY
BROOME
CATTARAUGUS
CAYUGA
CHAUTAUQUA
CHEMUNG
CHENANGO
CLINTON
COLUMBIA
CORTLAND
DELAWARE
DUTCHESS
ERIE
ESSEX
FRANKLIN
FULTON
GENESEE
GREENE
HAMILTON
HERKIMER
JEFFERSON
LEWIS
LIVINGSTON
MADISON
MONROE
MONTGOMERY
NASSAU
NEW YORK
BRONX
KINGS
QUEENS
RICHMOND
NIAGARA
ONEIDA
ONONDAGA
ONTARIO
ORANGE
ORLEANS
OSWEGO
OTSEGO
PUTNAM
RENSSELAER
ROCKLAND
SAINT LAWRENCE
SARATOGA
SCHENECTADY
SCHOHARIE
SCHUYLER
SENECA
STEUBEN
SUFFOLK
SULLIVAN
TIOGA
TOMPKINS
ULSTER
WARREN
WASHINGTON
WAYNE
WESTCHESTER
WYOMING
YATES
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68
county_code

dropout_pct

```
df["dropout_pct"].describe()
```

	dropout_pct
count	39674.000000
mean	7.963049
std	9.658698
min	0.000000
25%	1.000000
50%	5.000000
75%	11.000000
max	100.000000

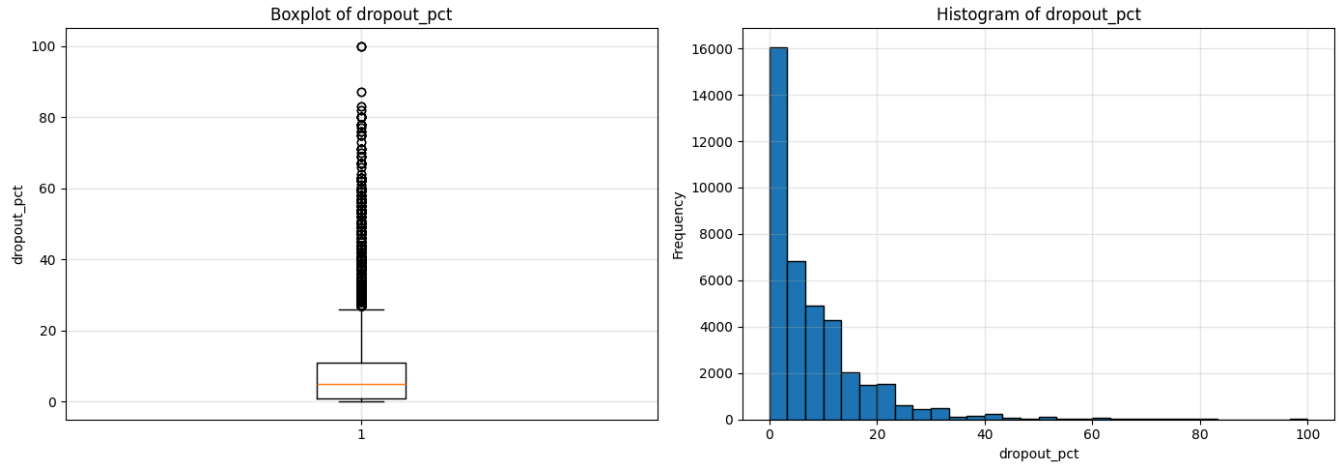
dtype: float64

```
# Create side-by-side plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Boxplot
axes[0].boxplot(df["dropout_pct"].dropna())
axes[0].set_ylabel('dropout_pct')
axes[0].set_title('Boxplot of dropout_pct')
axes[0].grid(True, alpha=0.3)

# Histogram
axes[1].hist(df["dropout_pct"].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('dropout_pct')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of dropout_pct')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



dropout_cnt

```
# Convert dropout_cnt to numeric (replace '-' with NaN)

df['dropout_cnt'].describe()
```

	dropout_cnt
count	39674.000000
mean	16.239225
std	50.129834
min	0.000000
25%	1.000000
50%	3.000000
75%	9.000000
max	1091.000000

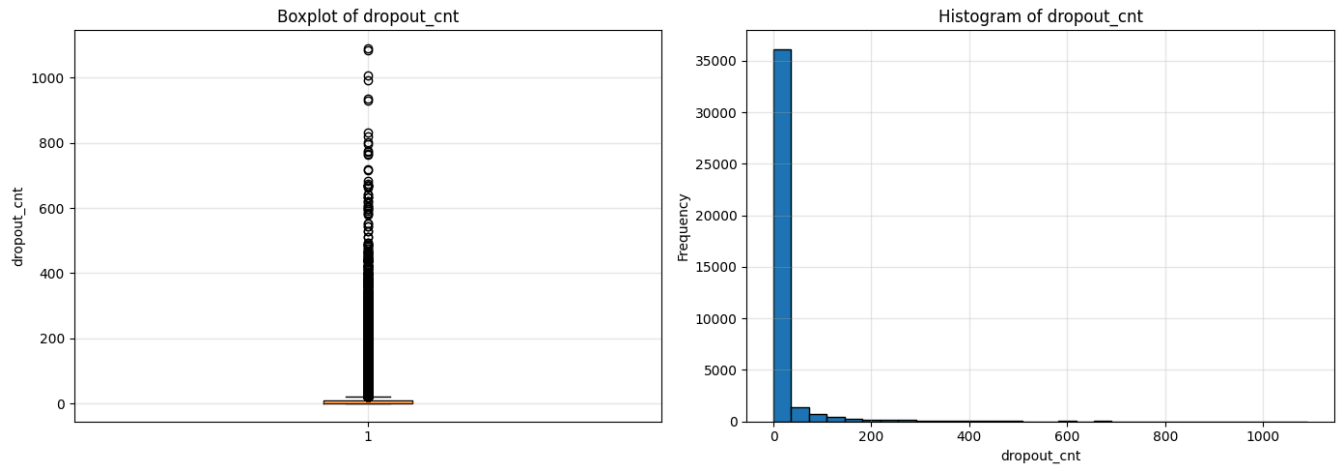
dtype: float64

```
# Create side-by-side plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Boxplot
axes[0].boxplot(df['dropout_cnt'].dropna())
axes[0].set_ylabel('dropout_cnt')
axes[0].set_title('Boxplot of dropout_cnt')
axes[0].grid(True, alpha=0.3)

# Histogram
axes[1].hist(df['dropout_cnt'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('dropout_cnt')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of dropout_cnt')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



ged_pct

```
#ged_pct
df['ged_pct'].describe()
```

	ged_pct
count	39674.000000
mean	0.612693
std	1.985445
min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	67.000000

dtype: float64

ged_cnt

```
#ged_cnt
df['ged_cnt'].describe()
```

	ged_cnt
count	39674.000000
mean	1.377577
std	4.949389
min	0.000000
25%	0.000000
50%	0.000000
75%	1.000000
max	97.000000

dtype: float64

still_enr_pct

```
#still_enr_pct
df['still_enr_pct'].describe()
```

	still_enr_pct
count	39674.000000
mean	5.190704
std	8.832710
min	0.000000
25%	0.000000
50%	2.000000
75%	6.000000
max	100.000000

dtype: float64

still_enr_cnt

```
#still_enr_cnt
df['still_enr_cnt'].describe()
```

	still_enr_cnt
count	39674.000000
mean	11.299516
std	40.766672
min	0.000000
25%	0.000000
50%	2.000000
75%	5.000000
max	1381.000000

dtype: float64

non_diploma_credential_pct

```
#non_diploma_credential_pct
df['non_diploma_credential_pct'].describe()
```

	non_diploma_credential_pct
count	39674.000000
mean	1.742627
std	4.063987
min	0.000000
25%	0.000000
50%	0.000000
75%	2.000000
max	67.000000

dtype: float64

non_diploma_credential_cnt

```
#non_diploma_credential_cnt
df['non_diploma_credential_cnt'].describe()
```

	non_diploma_credential_cnt
count	39674.000000
mean	1.924485
std	6.498913
min	0.000000
25%	0.000000
50%	0.000000
75%	2.000000
max	279.000000

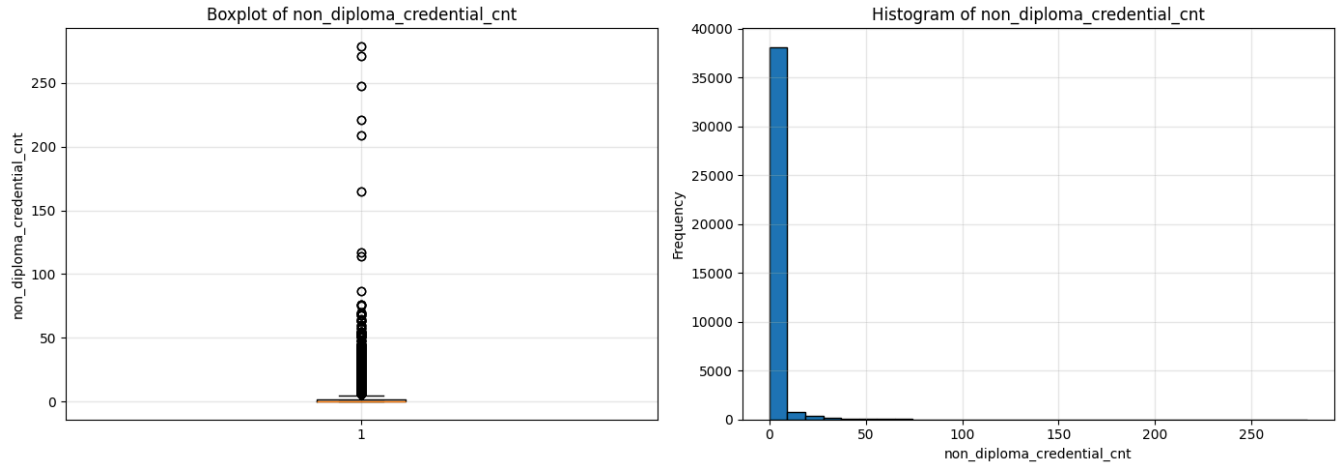
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['non_diploma_credential_cnt'].dropna())
axes[0].set_ylabel('non_diploma_credential_cnt')
axes[0].set_title('Boxplot of non_diploma_credential_cnt')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['non_diploma_credential_cnt'].dropna(), bins=30, edgecolor='black')
```

```
axes[1].set_xlabel('non_diploma_credential_cnt')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of non_diploma_credential_cnt')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



reg_avg_pct

```
#reg_adv_pct
df['reg_adv_pct'].describe()
```

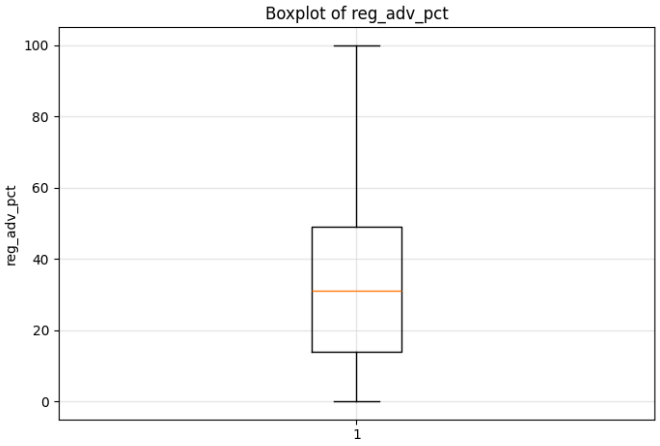
	reg_adv_pct
count	39674.000000
mean	32.577204
std	23.001197
min	0.000000
25%	14.000000
50%	31.000000
75%	49.000000
max	100.000000

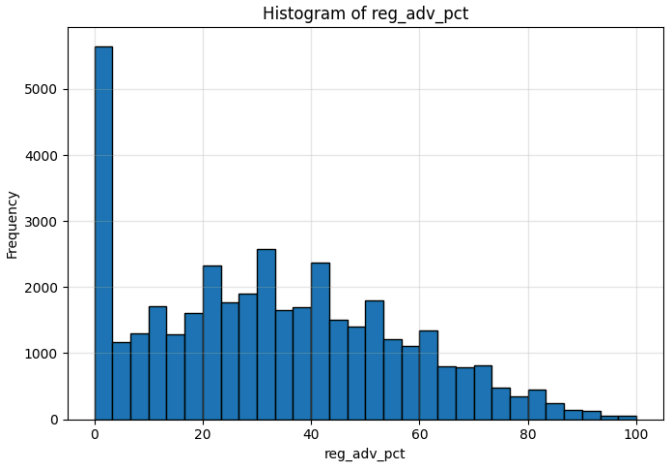
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['reg_adv_pct'].dropna())
axes[0].set_ylabel('reg_adv_pct')
axes[0].set_title('Boxplot of reg_adv_pct')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['reg_adv_pct'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('reg_adv_pct')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of reg_adv_pct')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





reg_avg_cnt

```
#reg_adv_cnt
df['reg_adv_cnt'].describe()
```

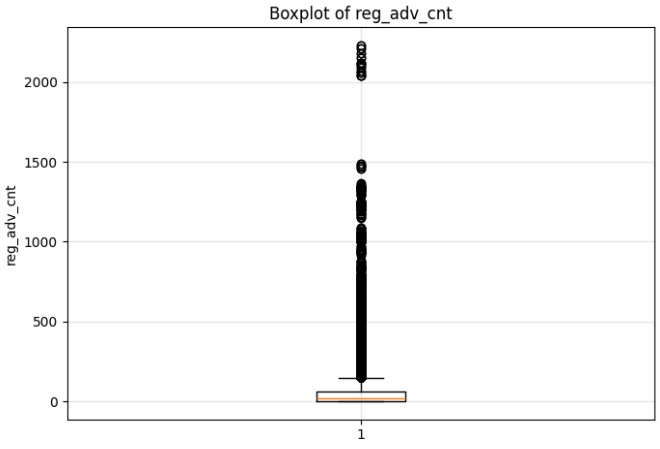
	reg_adv_cnt
count	39674.000000
mean	62.032742
std	132.777866
min	0.000000
25%	4.000000
50%	18.000000
75%	62.000000
max	2231.000000

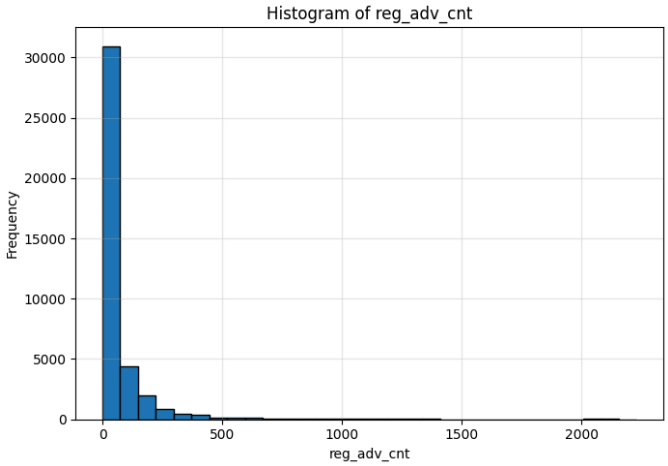
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['reg_adv_cnt'].dropna())
axes[0].set_ylabel('reg_adv_cnt')
axes[0].set_title('Boxplot of reg_adv_cnt')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['reg_adv_cnt'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('reg_adv_cnt')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of reg_adv_cnt')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





reg_pct

```
#reg_pct
df['reg_pct'].describe()
```

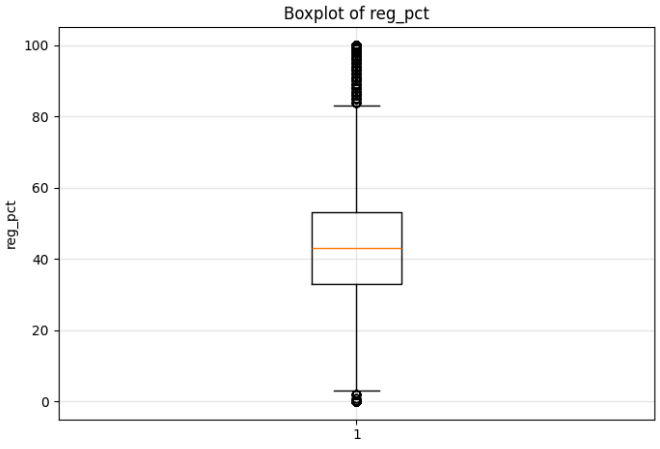
	reg_pct
count	39674.000000
mean	43.371125
std	17.124891
min	0.000000
25%	33.000000
50%	43.000000
75%	53.000000
max	100.000000

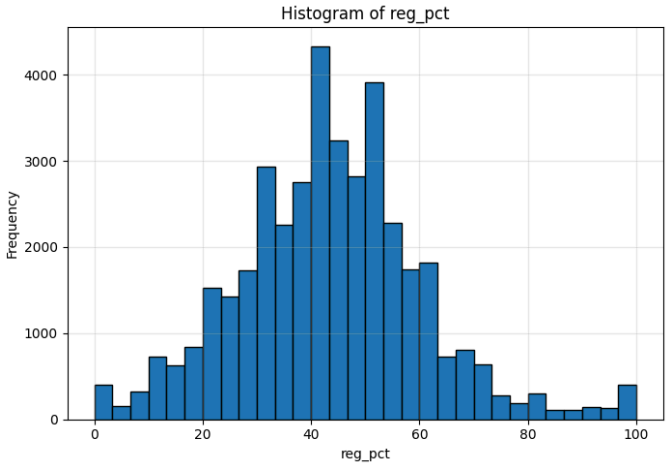
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['reg_pct'].dropna())
axes[0].set_ylabel('reg_pct')
axes[0].set_title('Boxplot of reg_pct')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['reg_pct'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('reg_pct')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of reg_pct')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





reg_cnt

```
#reg_cnt
df['reg_cnt'].describe()
```

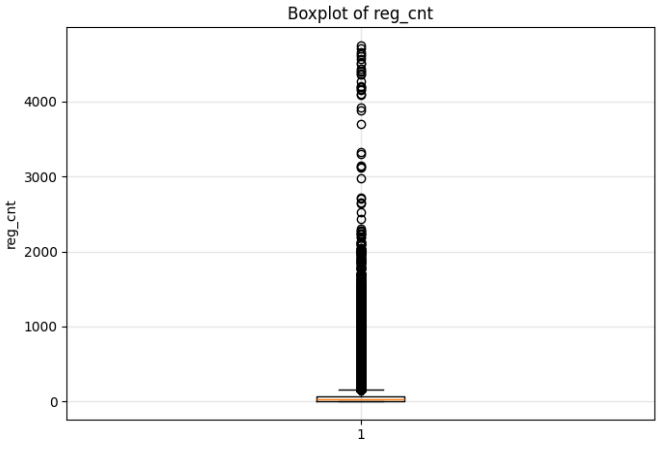
	reg_cnt
count	39674.000000
mean	86.804708
std	225.795826
min	0.000000
25%	10.000000
50%	27.000000
75%	69.000000
max	4752.000000

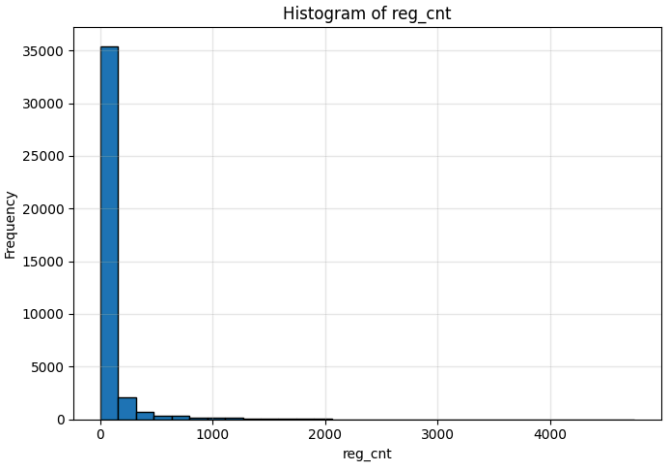
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['reg_cnt'].dropna())
axes[0].set_ylabel('reg_cnt')
axes[0].set_title('Boxplot of reg_cnt')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['reg_cnt'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('reg_cnt')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of reg_cnt')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





local_pct

```
#local_pct
df['local_pct'].describe()
```

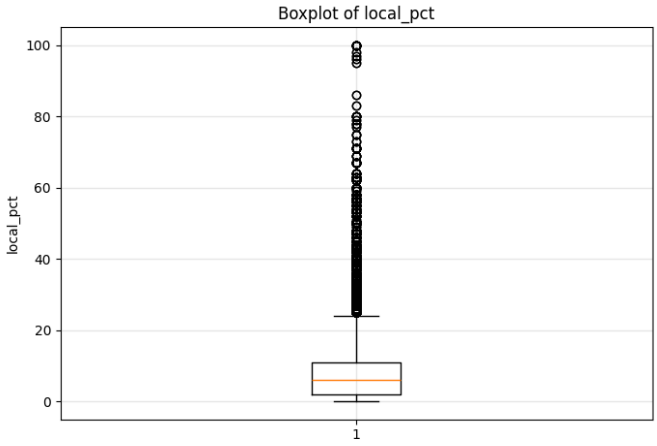
	local_pct
count	39674.000000
mean	8.479936
std	10.398486
min	0.000000
25%	2.000000
50%	6.000000
75%	11.000000
max	100.000000

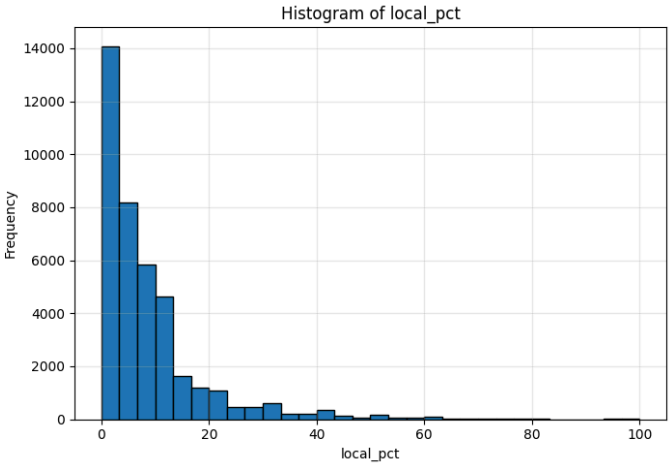
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['local_pct'].dropna())
axes[0].set_ylabel('local_pct')
axes[0].set_title('Boxplot of local_pct')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['local_pct'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('local_pct')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of local_pct')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





local_cnt

```
#local_cnt
df['local_cnt'].describe()
```

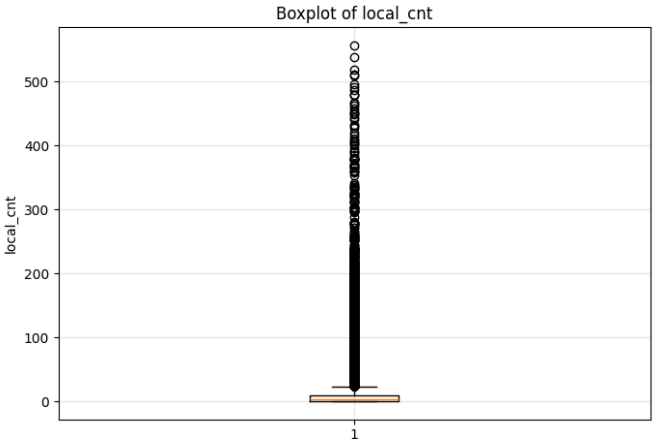
	local_cnt
count	39674.000000
mean	12.340903
std	32.046302
min	0.000000
25%	1.000000
50%	4.000000
75%	10.000000
max	557.000000

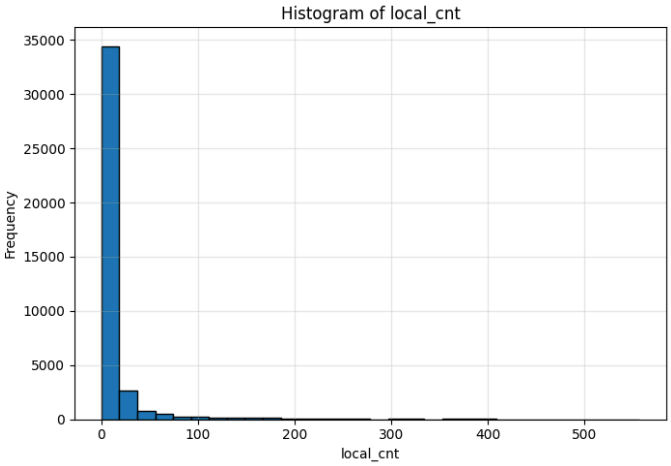
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['local_cnt'].dropna())
axes[0].set_ylabel('local_cnt')
axes[0].set_title('Boxplot of local_cnt')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['local_cnt'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('local_cnt')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of local_cnt')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





grad_pct

```
#grad_pct
df['grad_pct'].describe()
```

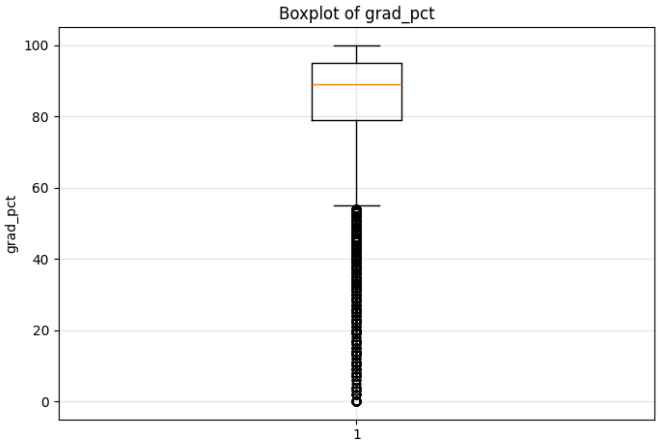
	grad_pct
count	39674.000000
mean	84.406614
std	15.679500
min	0.000000
25%	79.000000
50%	89.000000
75%	95.000000
max	100.000000

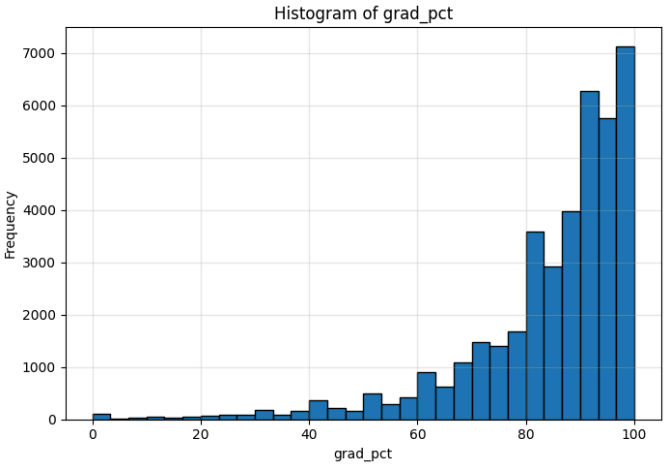
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['grad_pct'].dropna())
axes[0].set_ylabel('grad_pct')
axes[0].set_title('Boxplot of grad_pct')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['grad_pct'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('grad_pct')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of grad_pct')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





grad_cnt

```
#grad_cnt
df['grad_cnt'].describe()
```

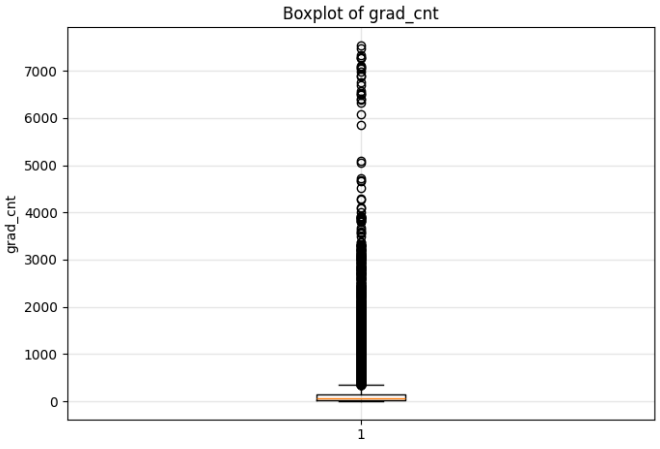
	grad_cnt
count	39674.000000
mean	161.178354
std	361.294773
min	0.000000
25%	20.000000
50%	57.000000
75%	156.000000
max	7540.000000

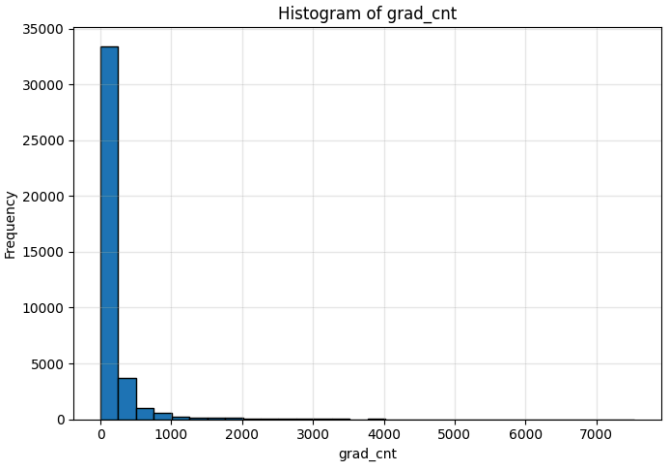
dtype: float64

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(df['grad_cnt'].dropna())
axes[0].set_ylabel('grad_cnt')
axes[0].set_title('Boxplot of grad_cnt')
axes[0].grid(True, alpha=0.3)

axes[1].hist(df['grad_cnt'].dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('grad_cnt')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of grad_cnt')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```





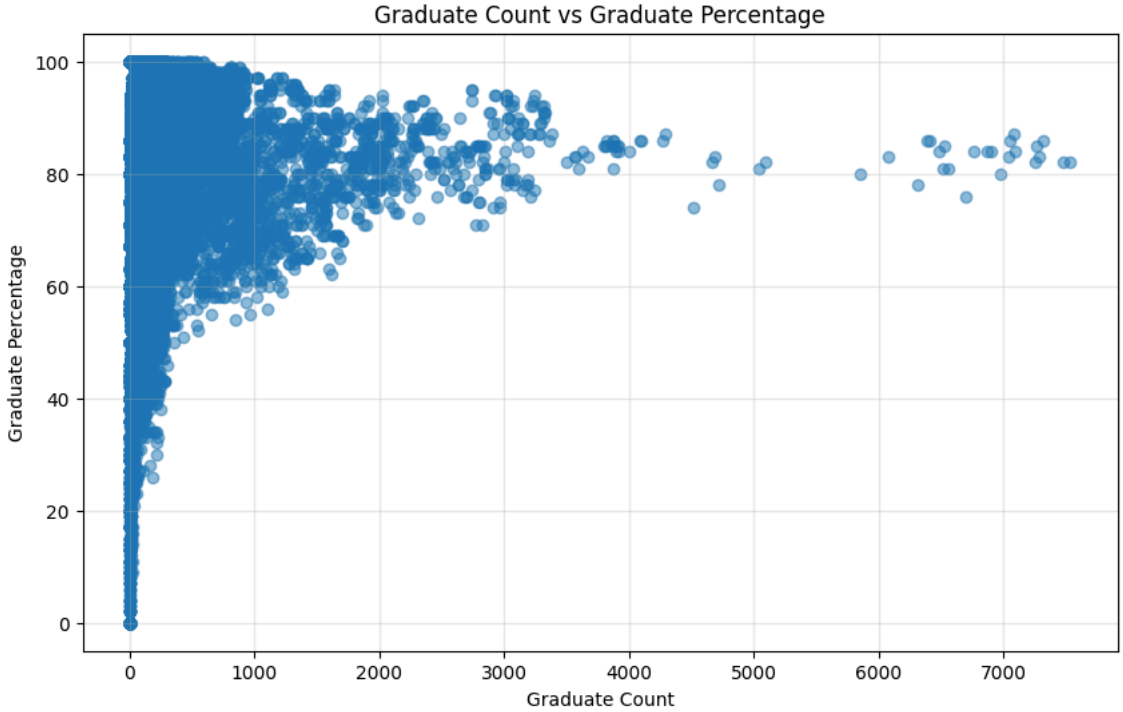
grad_pct & grad_cnt

```
# Scatter plot to see relationship
plt.figure(figsize=(10, 6))
plt.scatter(df['grad_cnt'], df['grad_pct'], alpha=0.5)
plt.xlabel('Graduate Count')
plt.ylabel('Graduate Percentage')
plt.title('Graduate Count vs Graduate Percentage')
plt.grid(True, alpha=0.3)
plt.show()

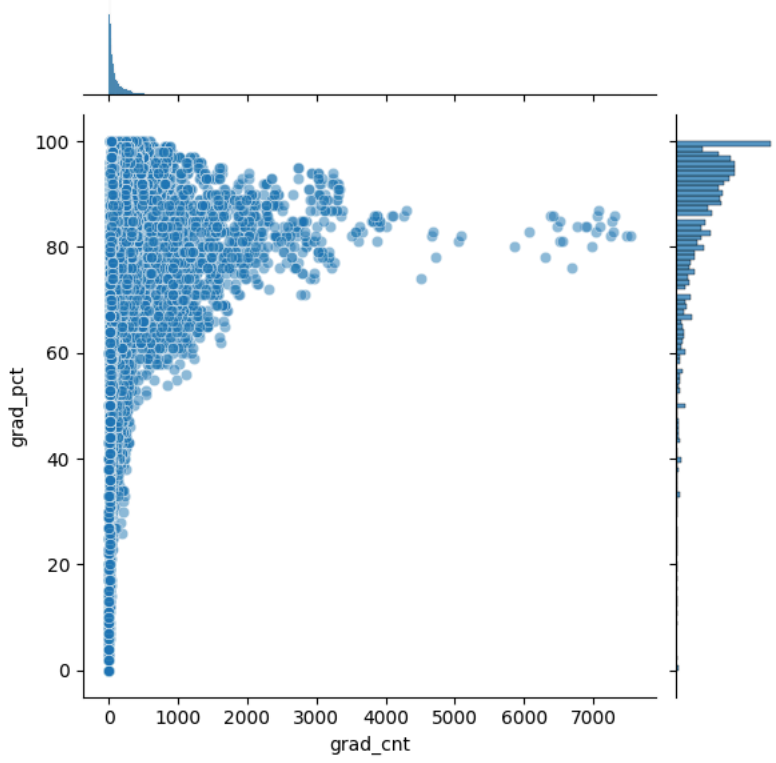
# Correlation
correlation = df[['grad_cnt', 'grad_pct']].corr()
print("Correlation Matrix:")
print(correlation)

# Joint plot with distributions
sns.jointplot(data=df, x='grad_cnt', y='grad_pct', kind='scatter', alpha=0.5)
plt.show()

# Summary statistics
print("\nSummary Statistics:")
print(df[['grad_cnt', 'grad_pct']].describe())
```




```
Correlation Matrix:
           grad_cnt  grad_pct
grad_cnt   1.000000  0.034515
grad_pct   0.034515  1.000000
```




```
Summary Statistics:
           grad_cnt      grad_pct
count  39674.000000  39674.000000
mean     161.178354     84.406614
std      361.294773     15.679500
min        0.000000      0.000000
25%       20.000000     79.000000
50%       57.000000     89.000000
75%      156.000000     95.000000
max     7540.000000    100.000000
```

enroll_cnt

```
#enroll_cnt
df['enroll_cnt'].describe()
```

	enroll_cnt
count	39674.000000
mean	192.120079
std	439.972474
min	5.000000
25%	25.000000
50%	66.000000
75%	179.000000
max	9176.000000

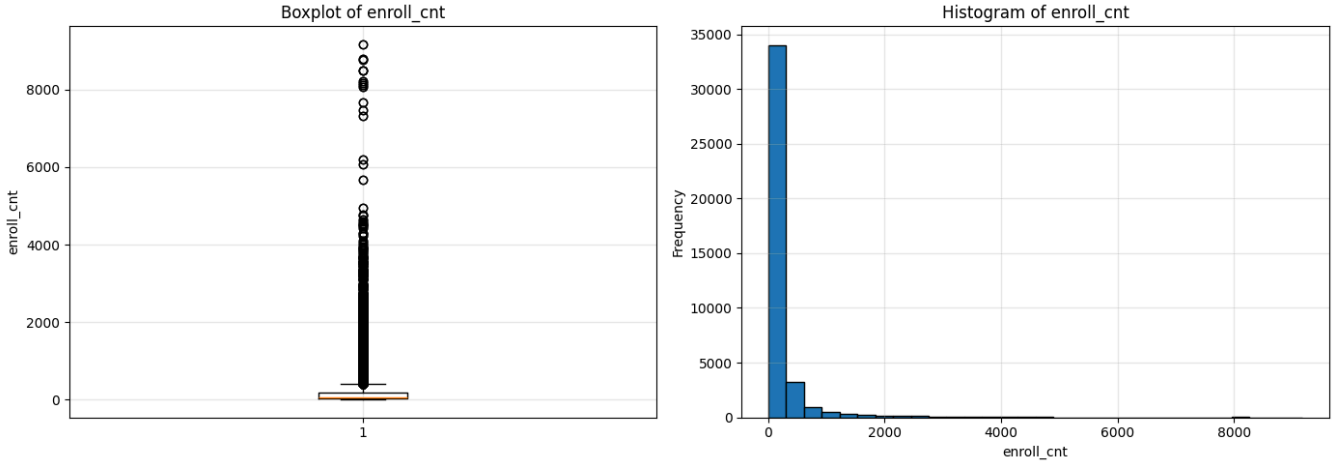
dtype: float64

```
# enroll_cnt
enroll_cnt_numeric = pd.to_numeric(df['enroll_cnt'].replace('-', None), errors='coerce')

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].boxplot(enroll_cnt_numeric.dropna())
axes[0].set_ylabel('enroll_cnt')
axes[0].set_title('Boxplot of enroll_cnt')
axes[0].grid(True, alpha=0.3)

axes[1].hist(enroll_cnt_numeric.dropna(), bins=30, edgecolor='black')
axes[1].set_xlabel('enroll_cnt')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Histogram of enroll_cnt')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



subgroup_name

```
#subgroup_name
df['subgroup_name'].describe()
```